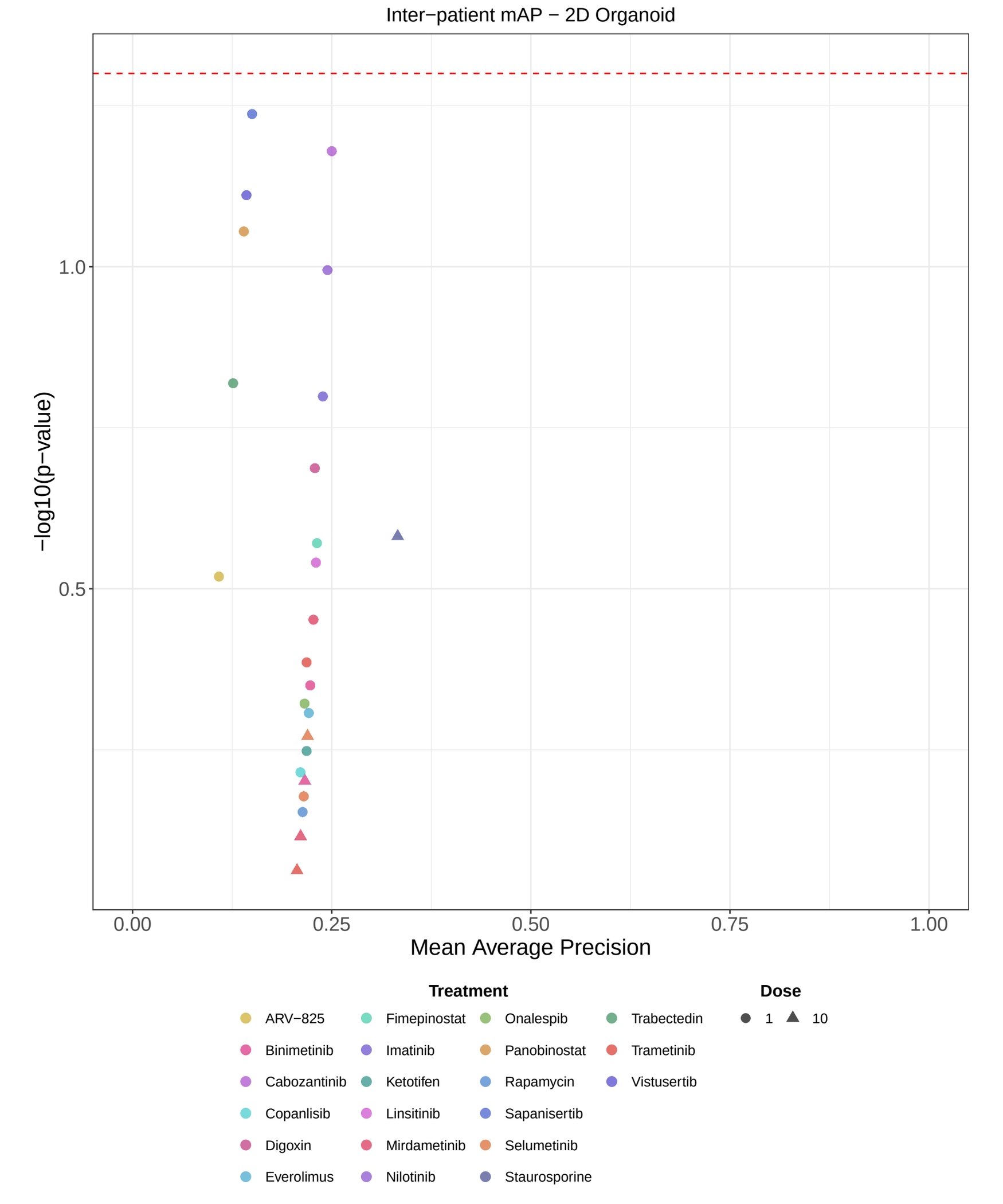
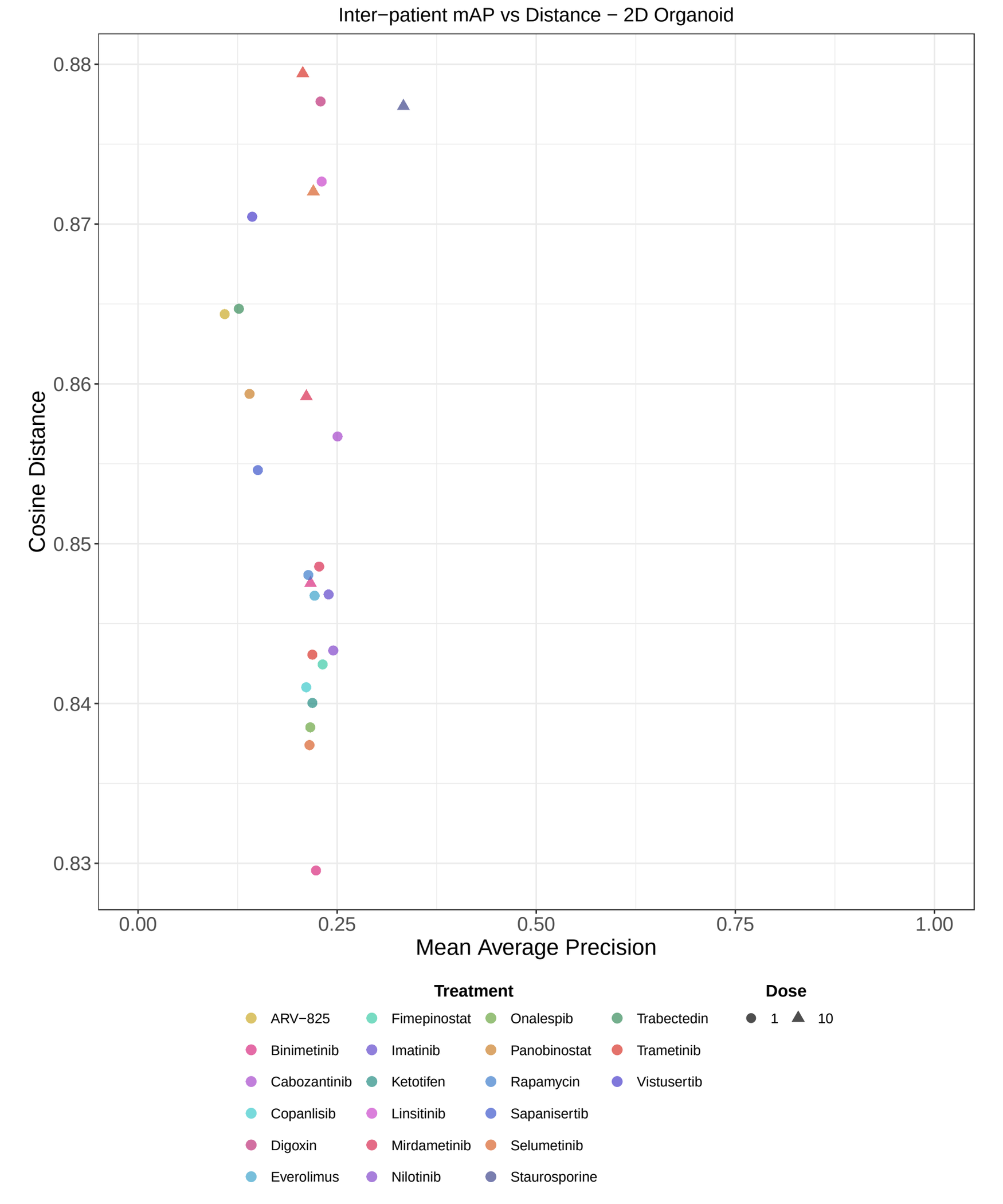
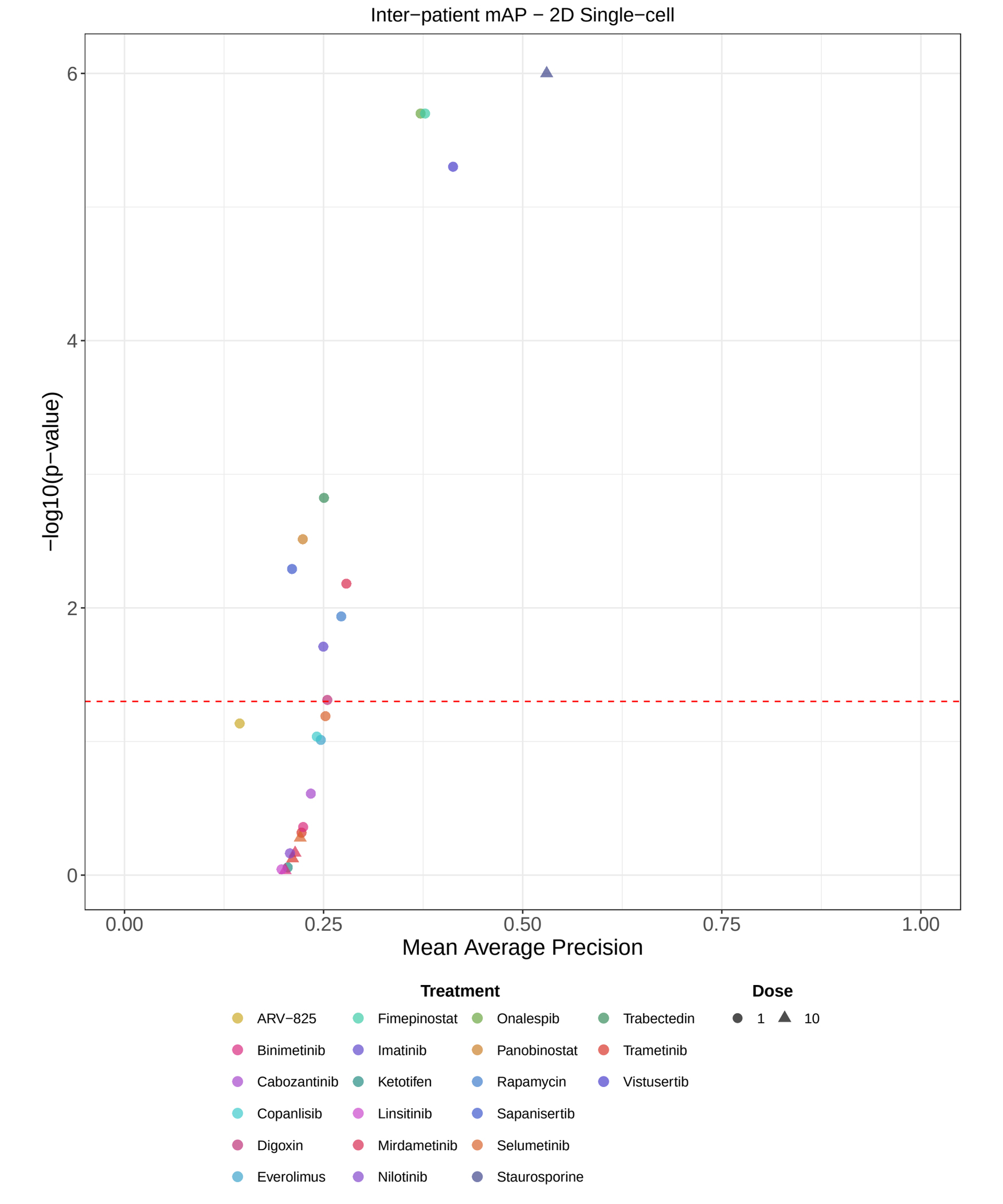
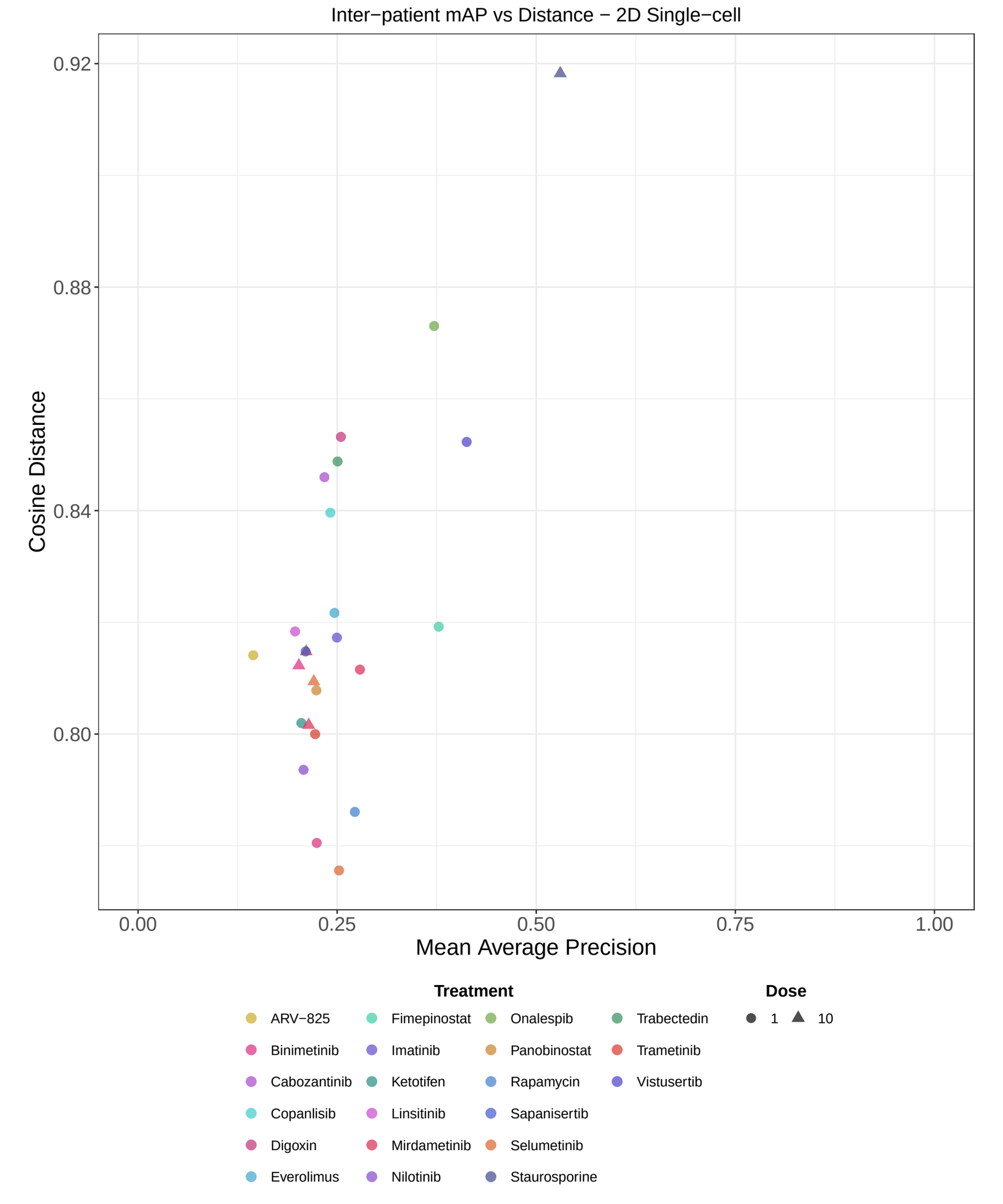
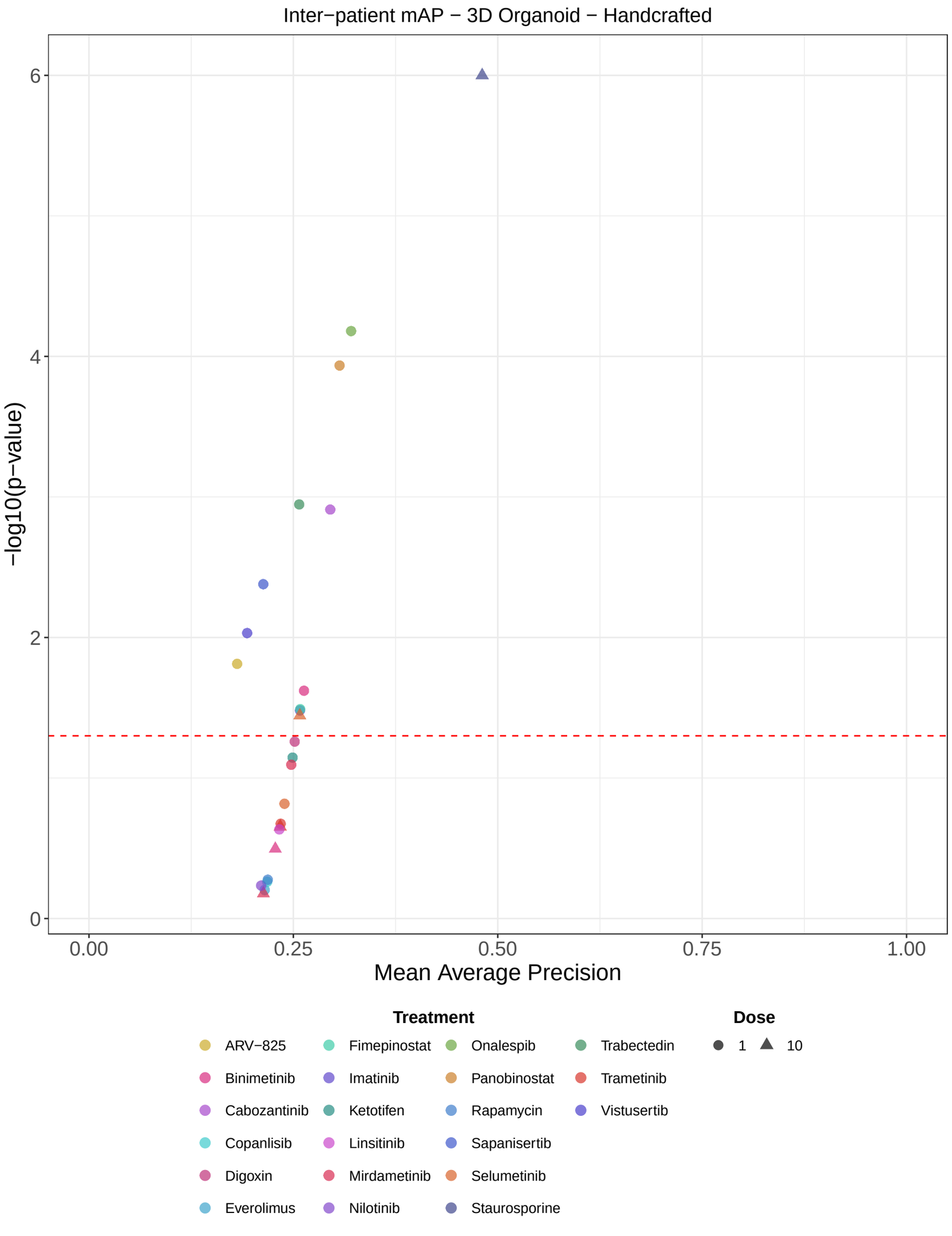
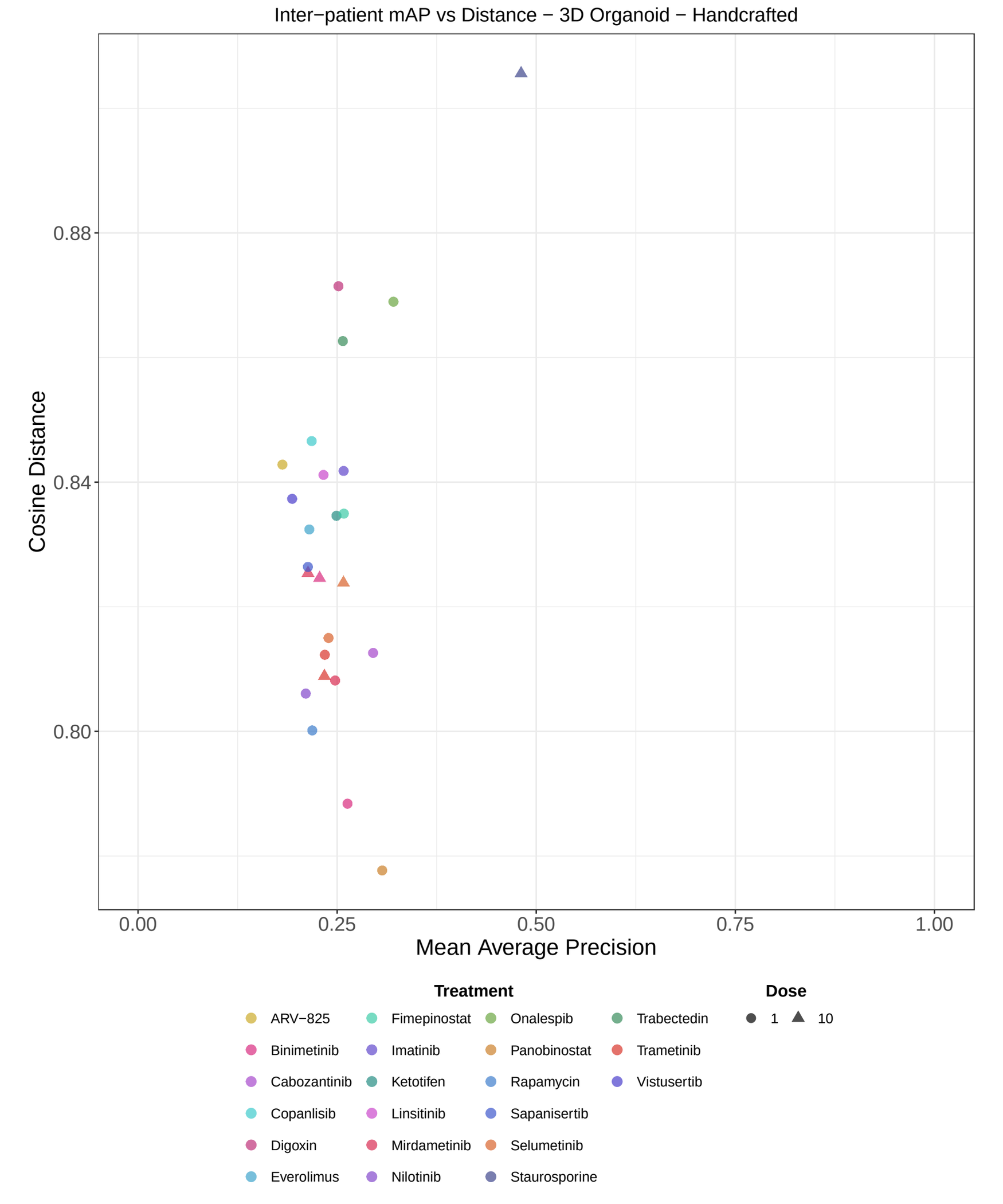

Inter−patient mAP − 2D Organoid
−log10(p−value)
1.0
0.5
0.00
0.25
0.50
0.75
1.00
Mean Average Precision
Treatment
ARV−825
Binimetinib
Cabozantinib
Copanlisib
Digoxin
Everolimus
Fimepinostat
Imatinib
Ketotifen
Linsitinib
Mirdametinib
Nilotinib
Onalespib
Panobinostat
Rapamycin
Sapanisertib
Selumetinib
Staurosporine
Trabectedin
Trametinib
Vistusertib
Dose
1
10

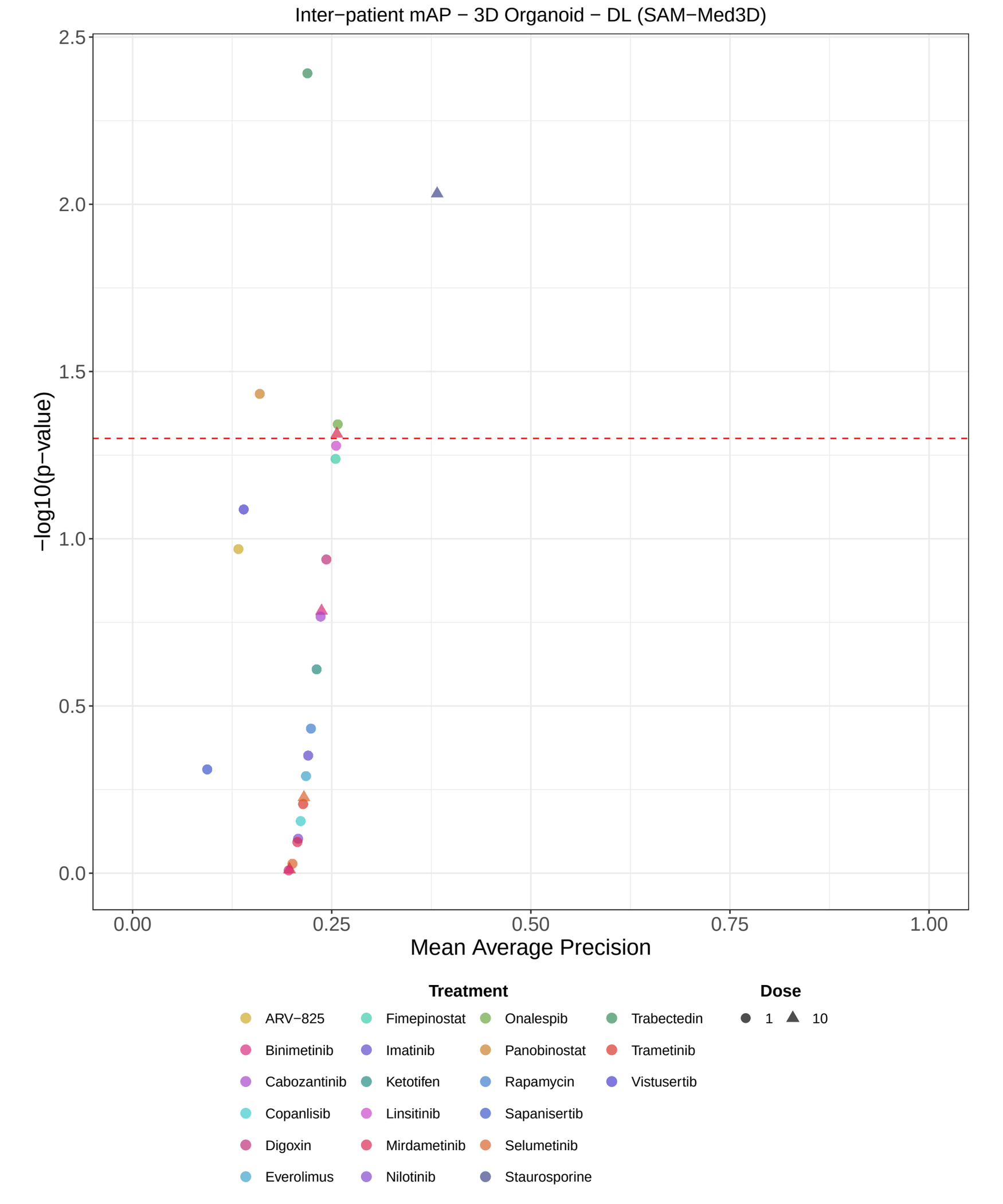

Inter−patient mAP vs Distance − 2D Organoid
Cosine Distance
0.88
0.87
0.86
0.85
0.84
0.83
0.00
0.25
0.50
0.75
1.00
Mean Average Precision
Treatment
ARV−825
Binimetinib
Cabozantinib
Copanlisib
Digoxin
Everolimus
Fimepinostat
Imatinib
Ketotifen
Linsitinib
Mirdametinib
Nilotinib
Onalespib
Panobinostat
Rapamycin
Sapanisertib
Selumetinib
Staurosporine
Trabectedin
Trametinib
Vistusertib
Dose
1
10

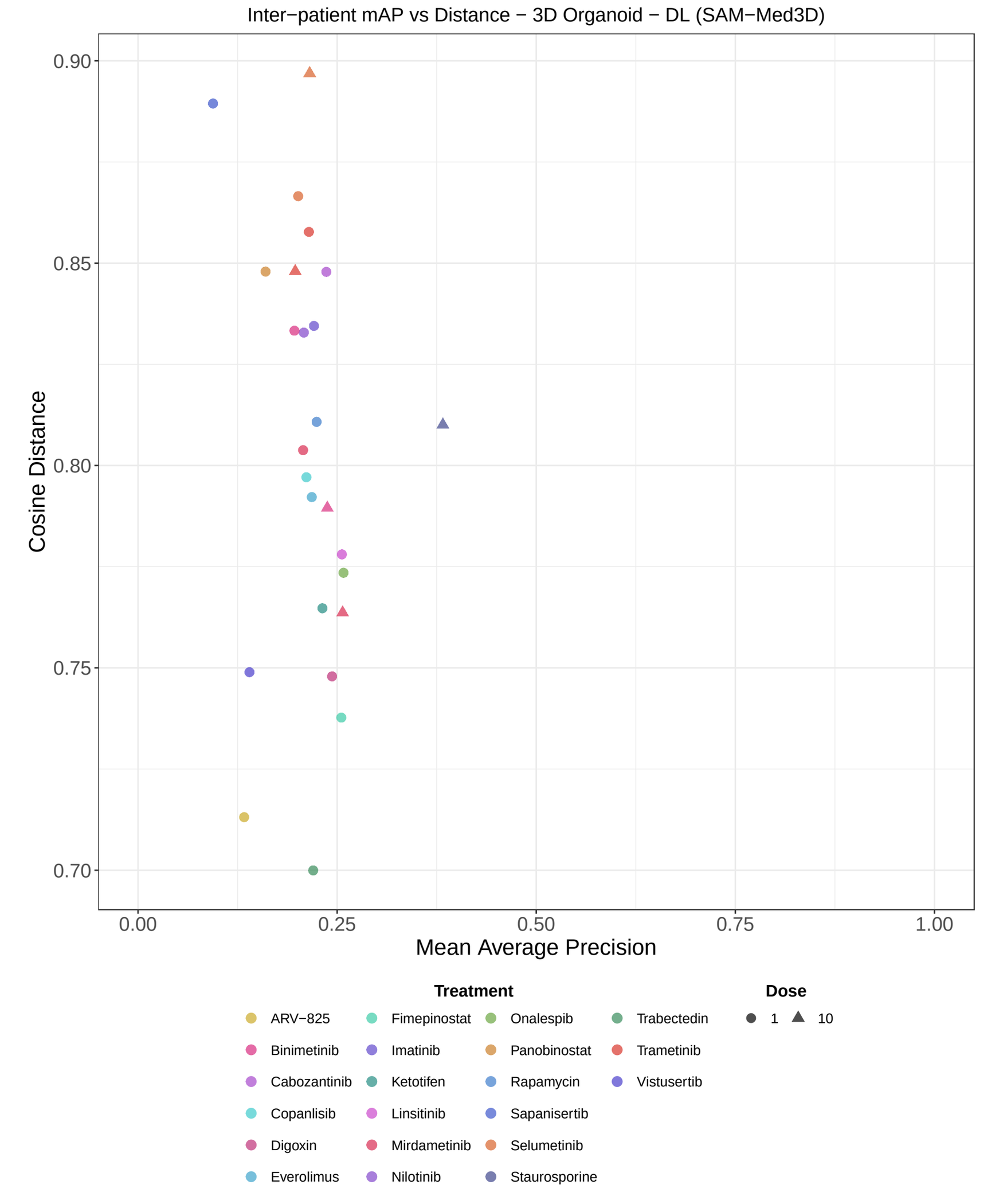

Inter−patient mAP − 2D Single−cell
−log10(p−value)
6
4
2
0
0.00
0.25
0.50
0.75
1.00
Mean Average Precision
Treatment
ARV−825
Binimetinib
Cabozantinib
Copanlisib
Digoxin
Everolimus
Fimepinostat
Imatinib
Ketotifen
Linsitinib
Mirdametinib
Nilotinib
Onalespib
Panobinostat
Rapamycin
Sapanisertib
Selumetinib
Staurosporine
Trabectedin
Trametinib
Vistusertib
Dose
1
10

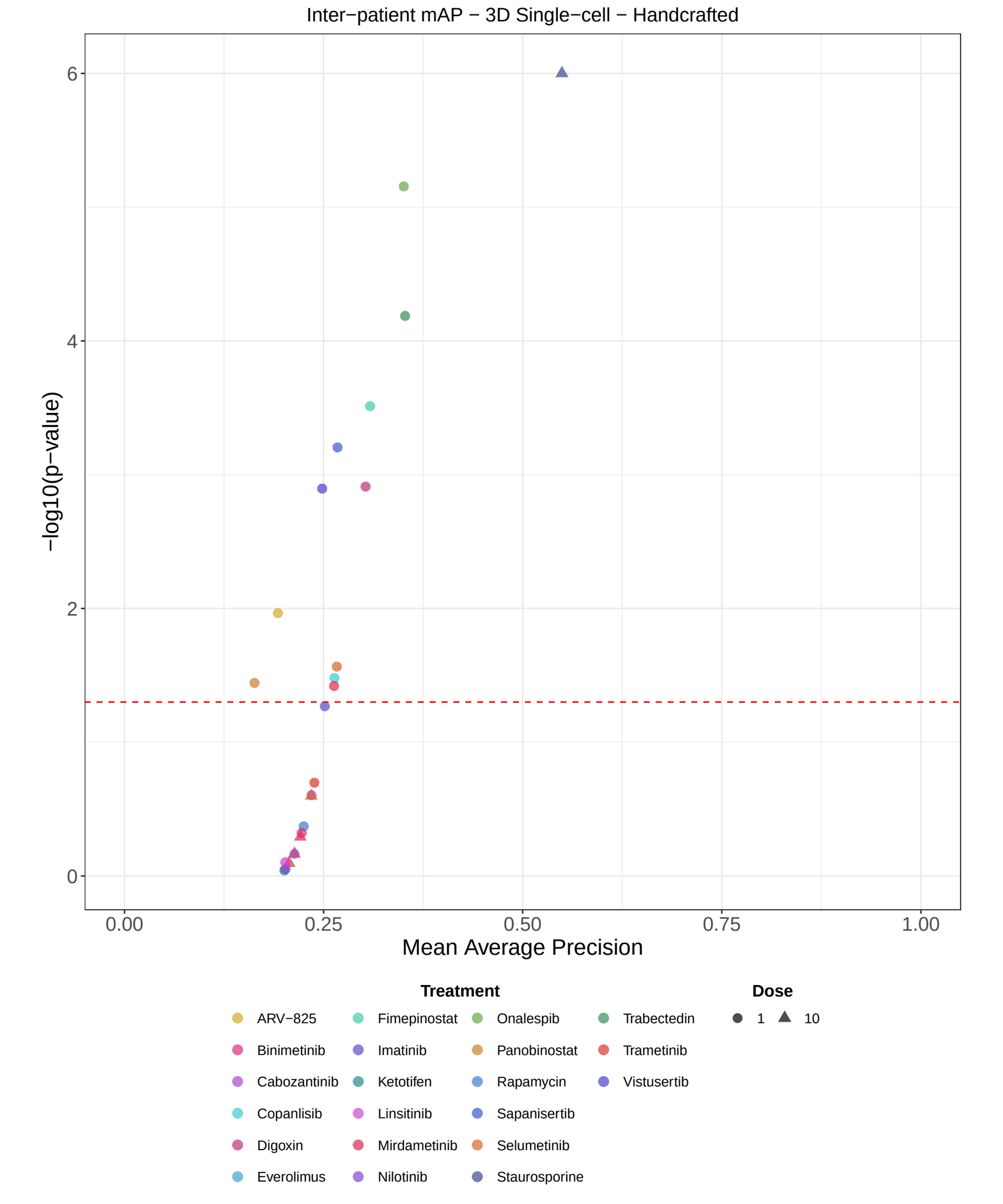

Inter−patient mAP vs Distance − 2D Single−cell
Cosine Distance
0.92
0.88
0.84
0.80
0.00
0.25
0.50
0.75
1.00
Mean Average Precision
Treatment
ARV−825
Binimetinib
Cabozantinib
Copanlisib
Digoxin
Everolimus
Fimepinostat
Imatinib
Ketotifen
Linsitinib
Mirdametinib
Nilotinib
Onalespib
Panobinostat
Rapamycin
Sapanisertib
Selumetinib
Staurosporine
Trabectedin
Trametinib
Vistusertib
Dose
1
10

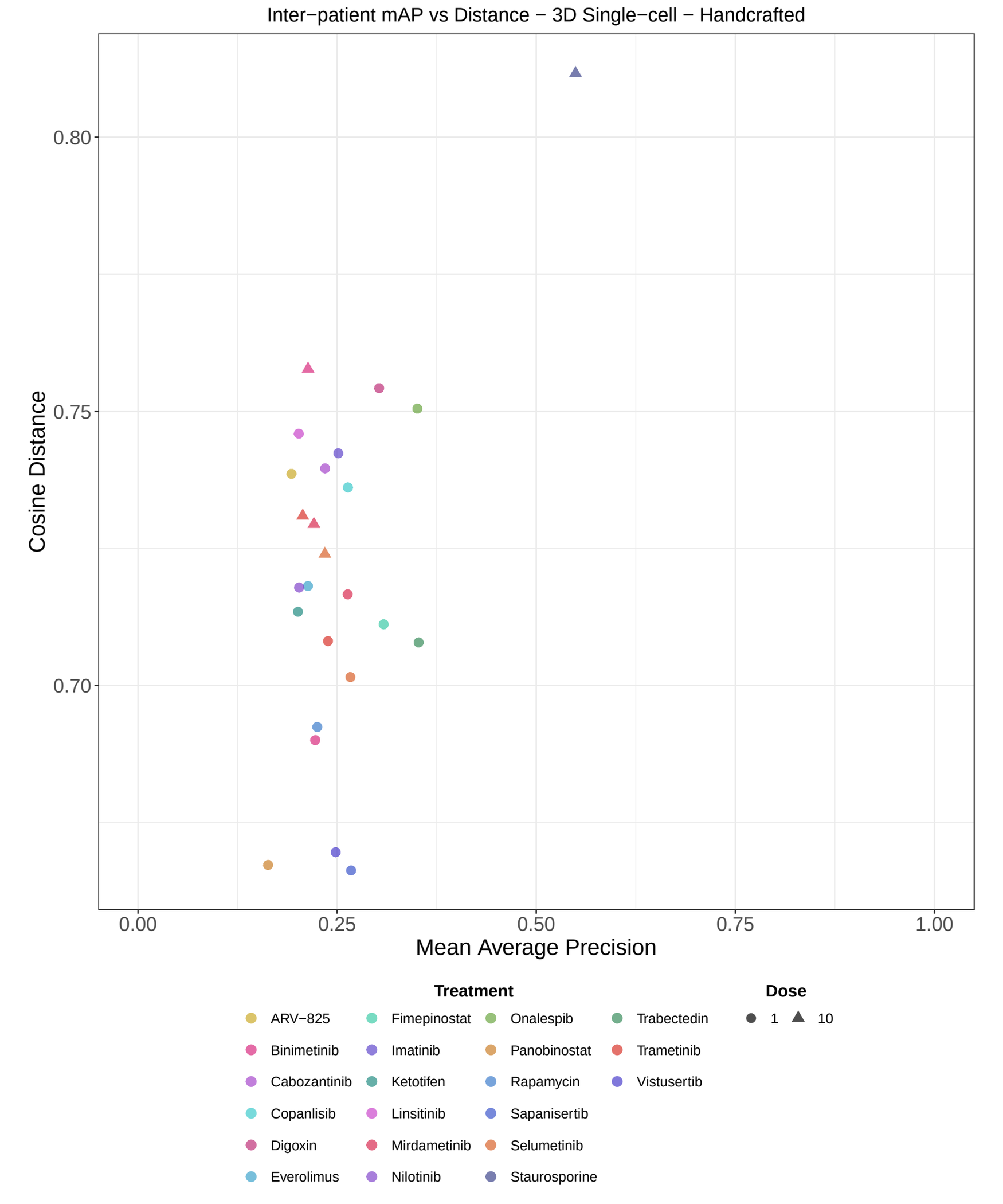

Inter−patient mAP − 3D Organoid − Handcrafted
−log10(p−value)
6
4
2
0
0.00
0.25
0.50
0.75
1.00
Mean Average Precision
Treatment
ARV−825
Binimetinib
Cabozantinib
Copanlisib
Digoxin
Everolimus
Fimepinostat
Imatinib
Ketotifen
Linsitinib
Mirdametinib
Nilotinib
Onalespib
Panobinostat
Rapamycin
Sapanisertib
Selumetinib
Staurosporine
Trabectedin
Trametinib
Vistusertib
Dose
1
10

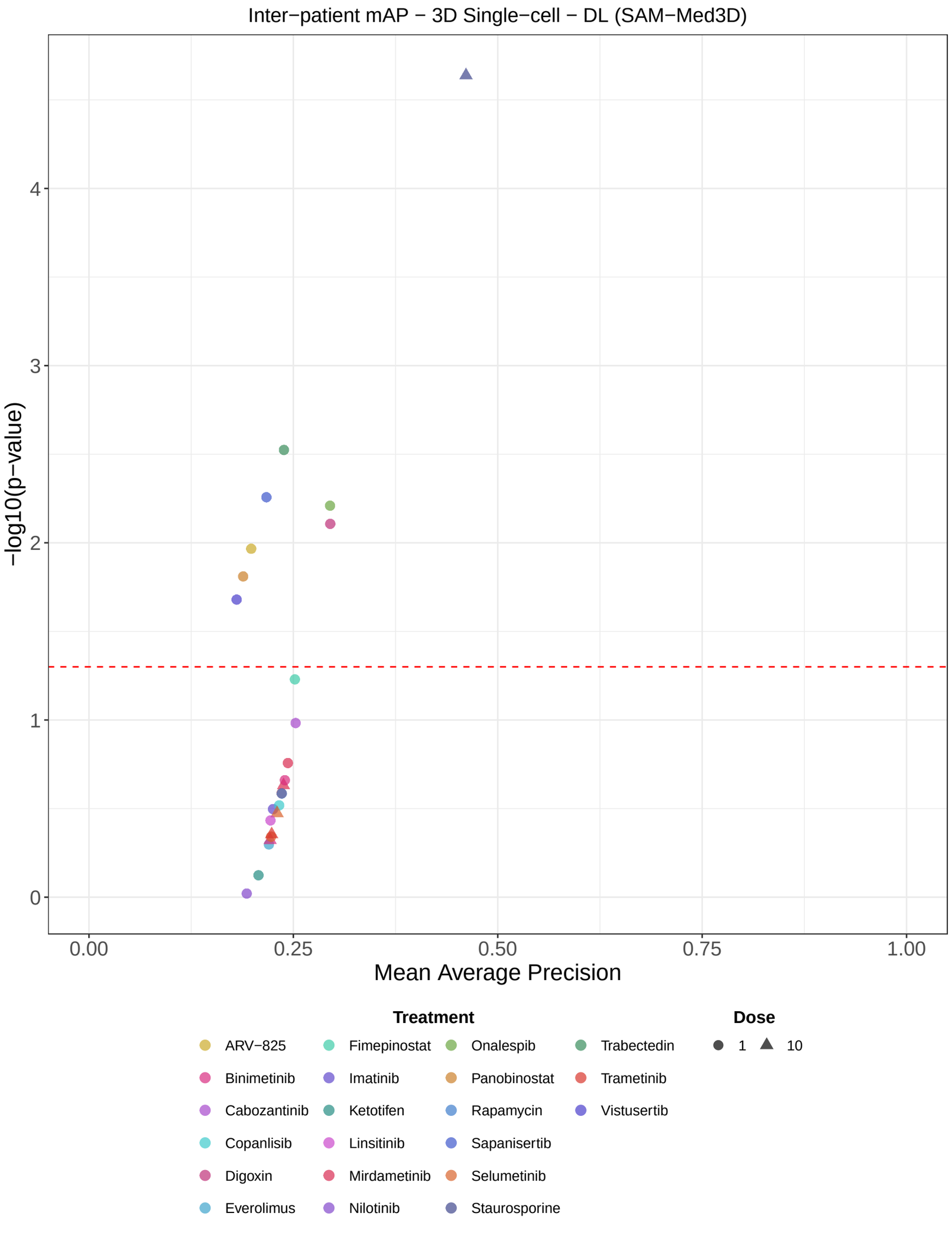

Inter−patient mAP vs Distance − 3D Organoid − Handcrafted
Cosine Distance
0.88
0.84
0.80
0.00
0.25
0.50
0.75
1.00
Mean Average Precision
Treatment
ARV−825
Binimetinib
Cabozantinib
Copanlisib
Digoxin
Everolimus
Fimepinostat
Imatinib
Ketotifen
Linsitinib
Mirdametinib
Nilotinib
Onalespib
Panobinostat
Rapamycin
Sapanisertib
Selumetinib
Staurosporine
Trabectedin
Trametinib
Vistusertib
Dose
1
10

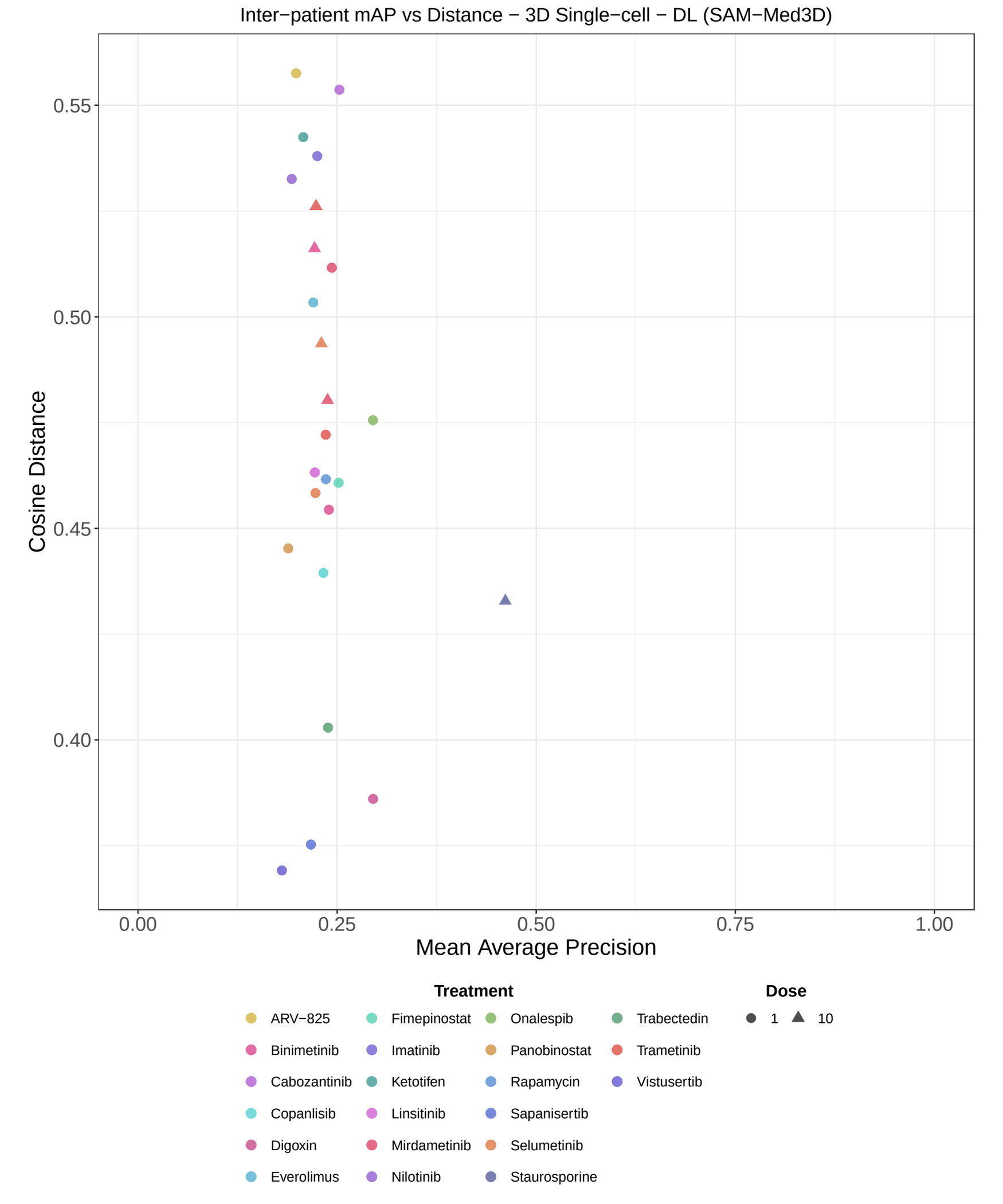

Inter−patient mAP − 3D Organoid − DL (SAM−Med3D)
−log10(p−value)
2.5
2.0
1.5
1.0
0.5
0.0
0.00
0.25
0.50
0.75
1.00
Mean Average Precision
Treatment
ARV−825
Binimetinib
Cabozantinib
Copanlisib
Digoxin
Everolimus
Fimepinostat
Imatinib
Ketotifen
Linsitinib
Mirdametinib
Nilotinib
Onalespib
Panobinostat
Rapamycin
Sapanisertib
Selumetinib
Staurosporine
Trabectedin
Trametinib
Vistusertib
Dose
1
10

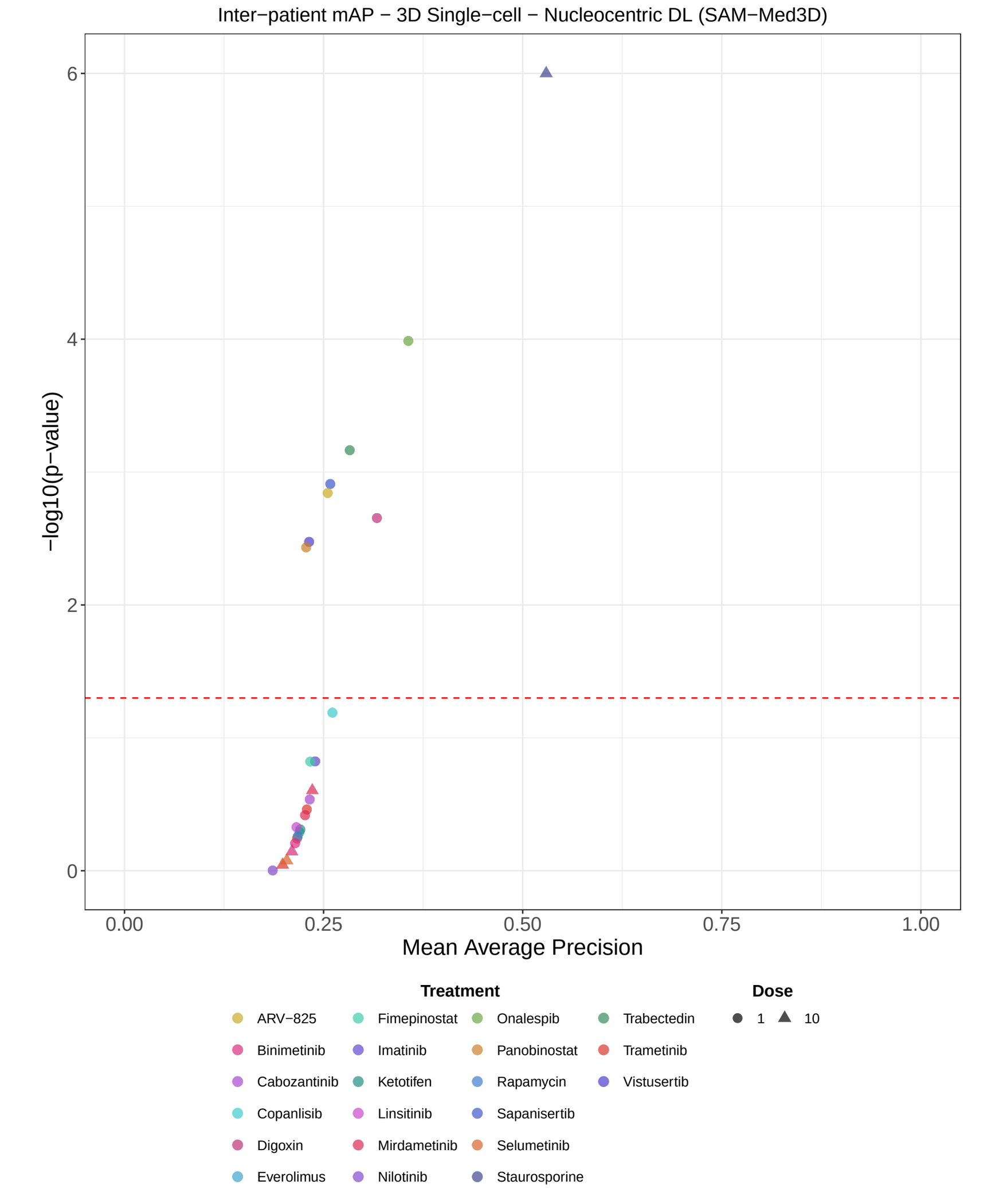

Inter−patient mAP vs Distance − 3D Organoid − DL (SAM−Med3D)
Cosine Distance
0.90
0.85
0.80
0.75
0.70
0.00
0.25
0.50
0.75
1.00
Mean Average Precision
Treatment
ARV−825
Binimetinib
Cabozantinib
Copanlisib
Digoxin
Everolimus
Fimepinostat
Imatinib
Ketotifen
Linsitinib
Mirdametinib
Nilotinib
Onalespib
Panobinostat
Rapamycin
Sapanisertib
Selumetinib
Staurosporine
Trabectedin
Trametinib
Vistusertib
Dose
1
10

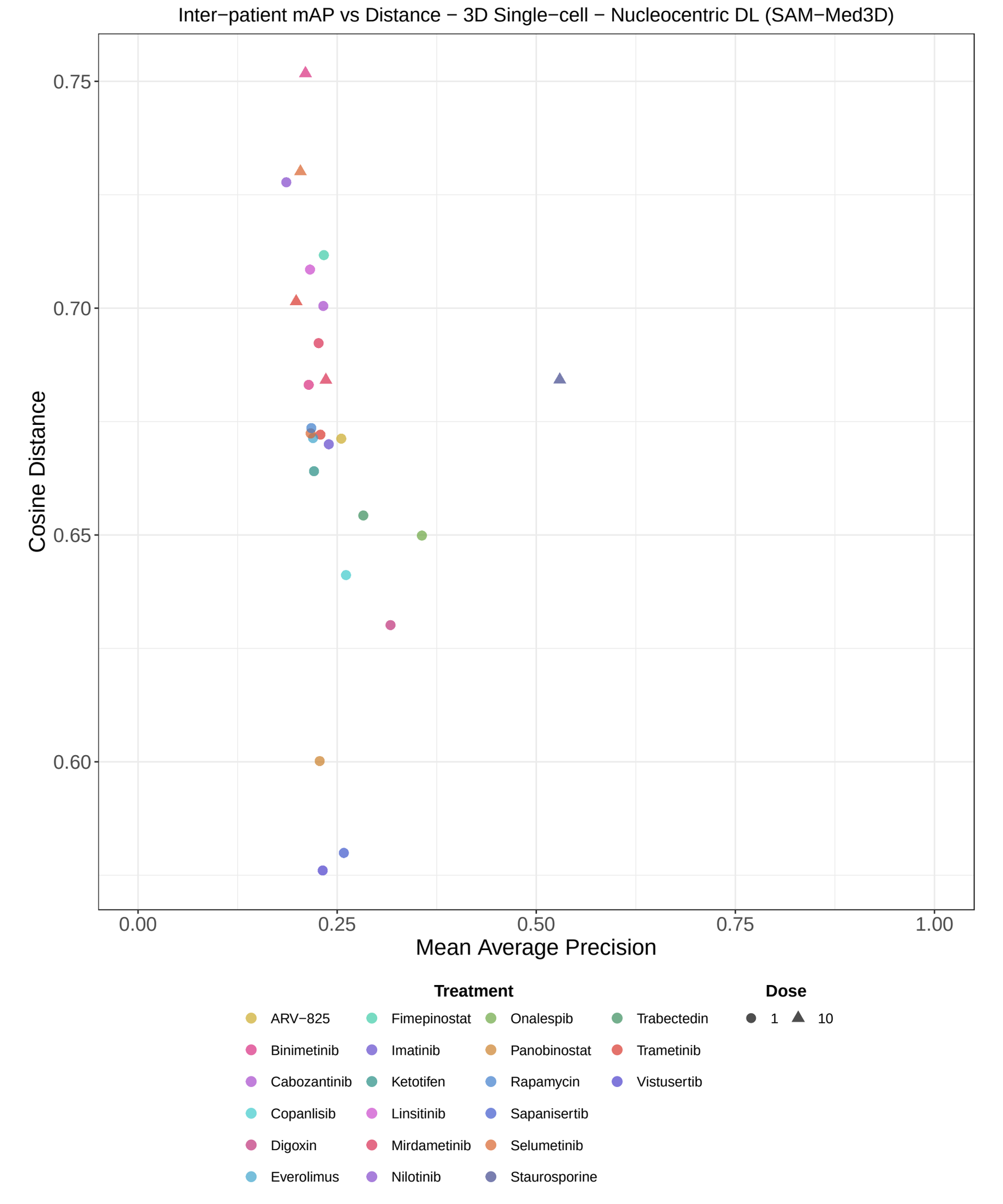

Inter−patient mAP − 3D Single−cell − Handcrafted
−log10(p−value)
6
4
2
0
0.00
0.25
0.50
0.75
1.00
Mean Average Precision
Treatment
ARV−825
Binimetinib
Cabozantinib
Copanlisib
Digoxin
Everolimus
Fimepinostat
Imatinib
Ketotifen
Linsitinib
Mirdametinib
Nilotinib
Onalespib
Panobinostat
Rapamycin
Sapanisertib
Selumetinib
Staurosporine
Trabectedin
Trametinib
Vistusertib
Dose
1
10

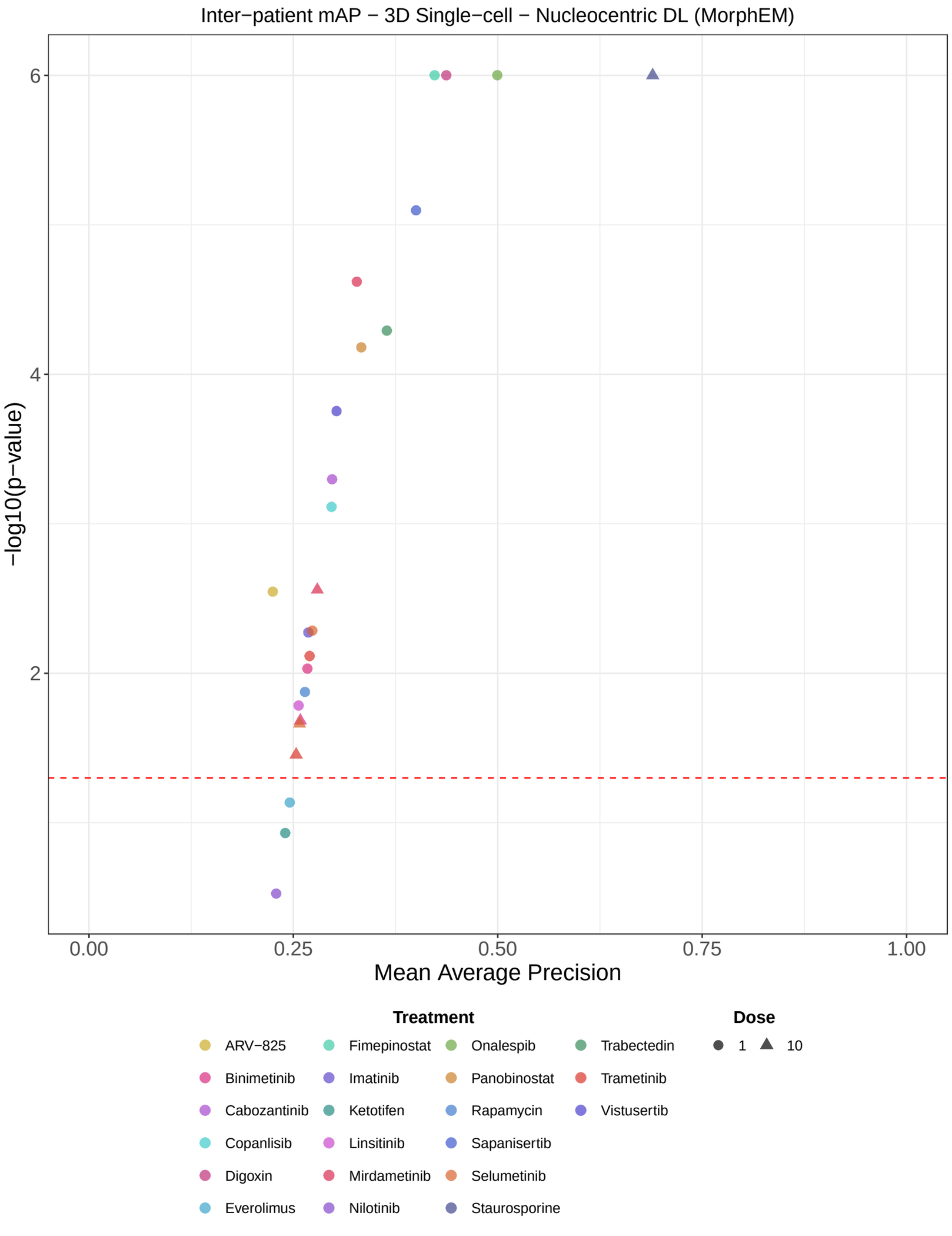

Inter−patient mAP vs Distance − 3D Single−cell − Handcrafted
Cosine Distance
0.80
0.75
0.70
0.00
0.25
0.50
0.75
1.00
Mean Average Precision
Treatment
ARV−825
Binimetinib
Cabozantinib
Copanlisib
Digoxin
Everolimus
Fimepinostat
Imatinib
Ketotifen
Linsitinib
Mirdametinib
Nilotinib
Onalespib
Panobinostat
Rapamycin
Sapanisertib
Selumetinib
Staurosporine
Trabectedin
Trametinib
Vistusertib
Dose
1
10

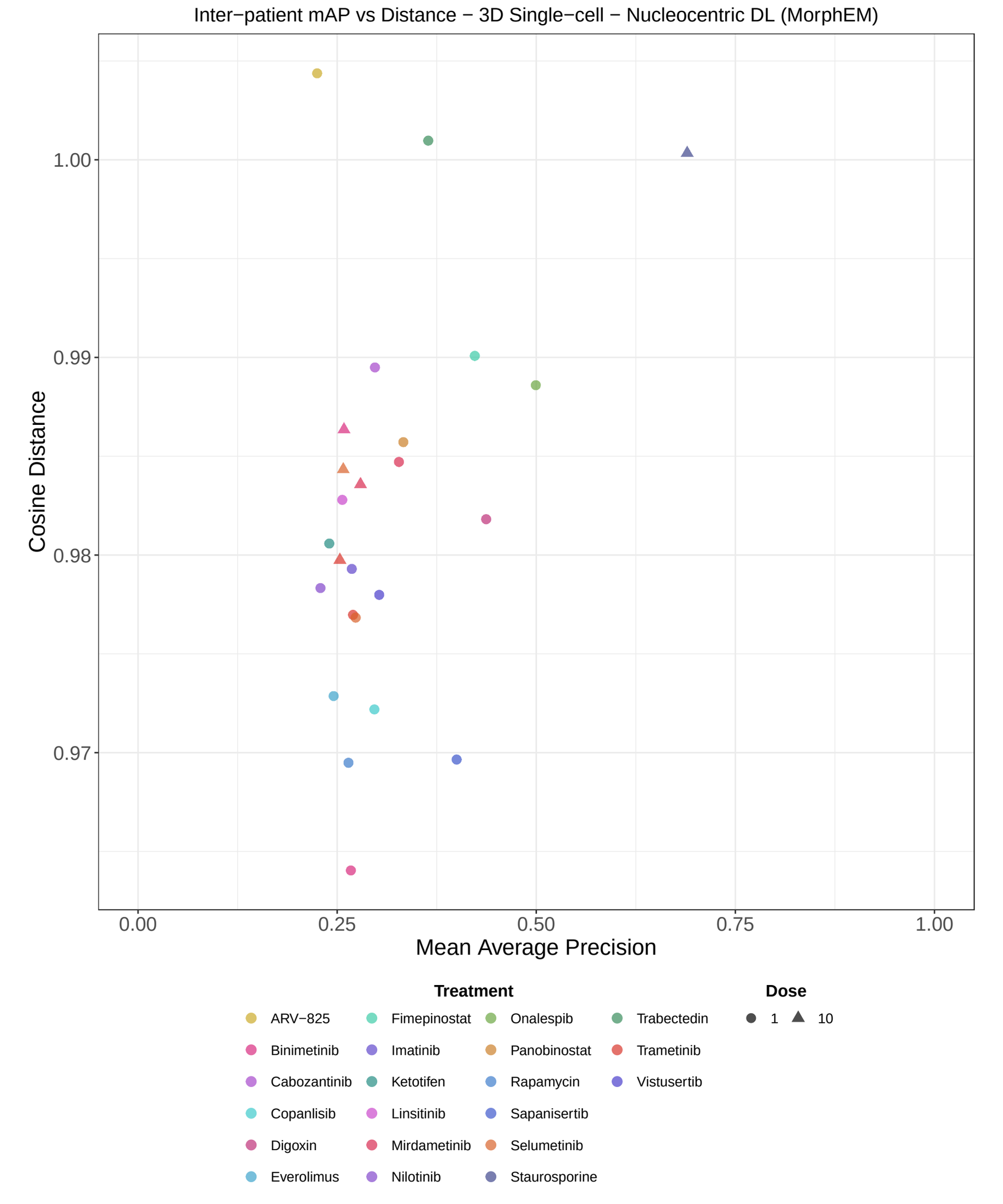

Inter−patient mAP − 3D Single−cell − DL (SAM−Med3D)
−log10(p−value)
0
1
2
3
4
0.00
0.25
0.50
0.75
1.00
Mean Average Precision
Treatment
ARV−825
Binimetinib
Cabozantinib
Copanlisib
Digoxin
Everolimus
Fimepinostat
Imatinib
Ketotifen
Linsitinib
Mirdametinib
Nilotinib
Onalespib
Panobinostat
Rapamycin
Sapanisertib
Selumetinib
Staurosporine
Trabectedin
Trametinib
Vistusertib
Dose
1
10

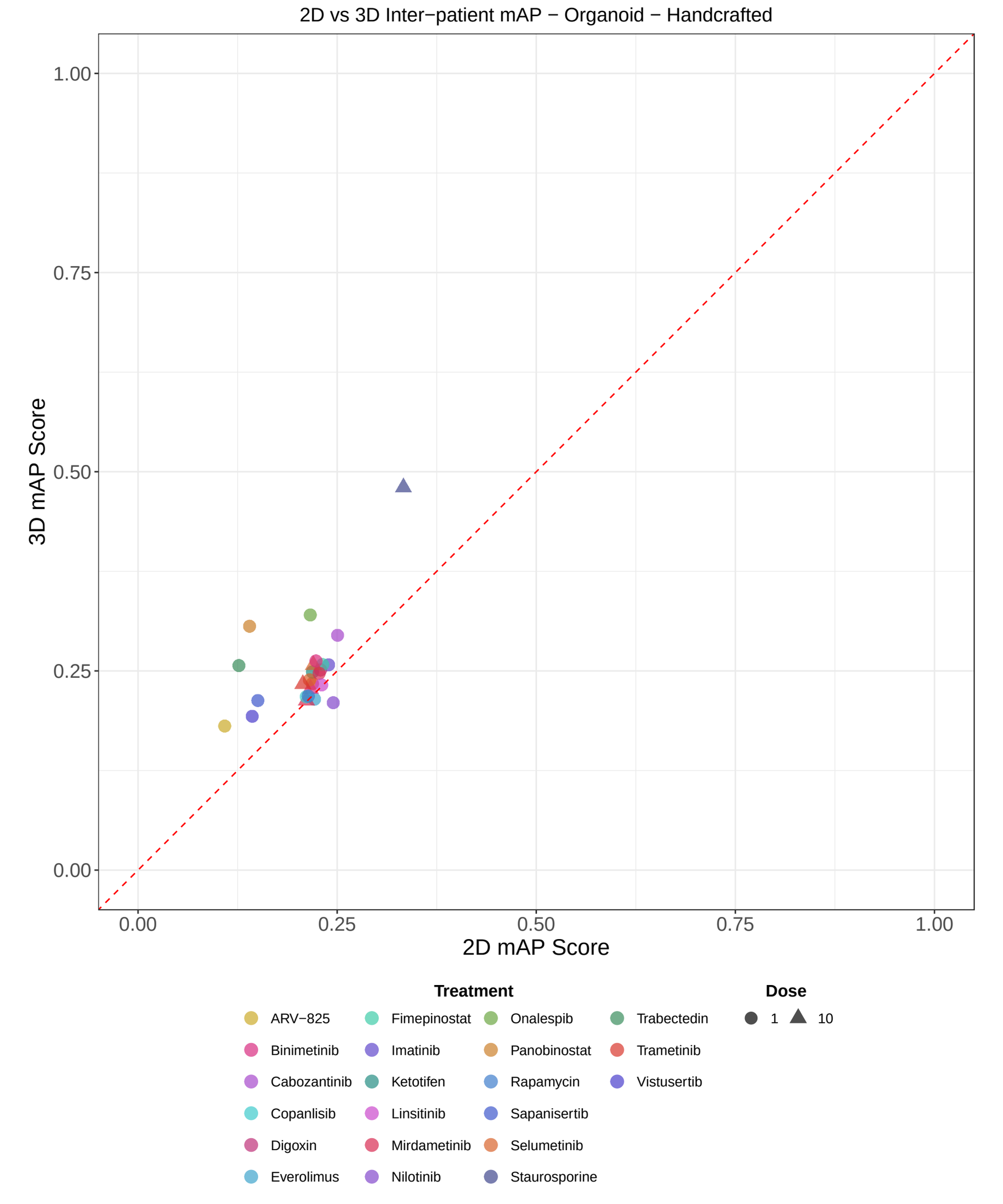

Inter−patient mAP vs Distance − 3D Single−cell − DL (SAM−Med3D)
Cosine Distance
0.55
0.50
0.45
0.40
0.00
0.25
0.50
0.75
1.00
Mean Average Precision
Treatment
ARV−825
Binimetinib
Cabozantinib
Copanlisib
Digoxin
Everolimus
Fimepinostat
Imatinib
Ketotifen
Linsitinib
Mirdametinib
Nilotinib
Onalespib
Panobinostat
Rapamycin
Sapanisertib
Selumetinib
Staurosporine
Trabectedin
Trametinib
Vistusertib
Dose
1
10

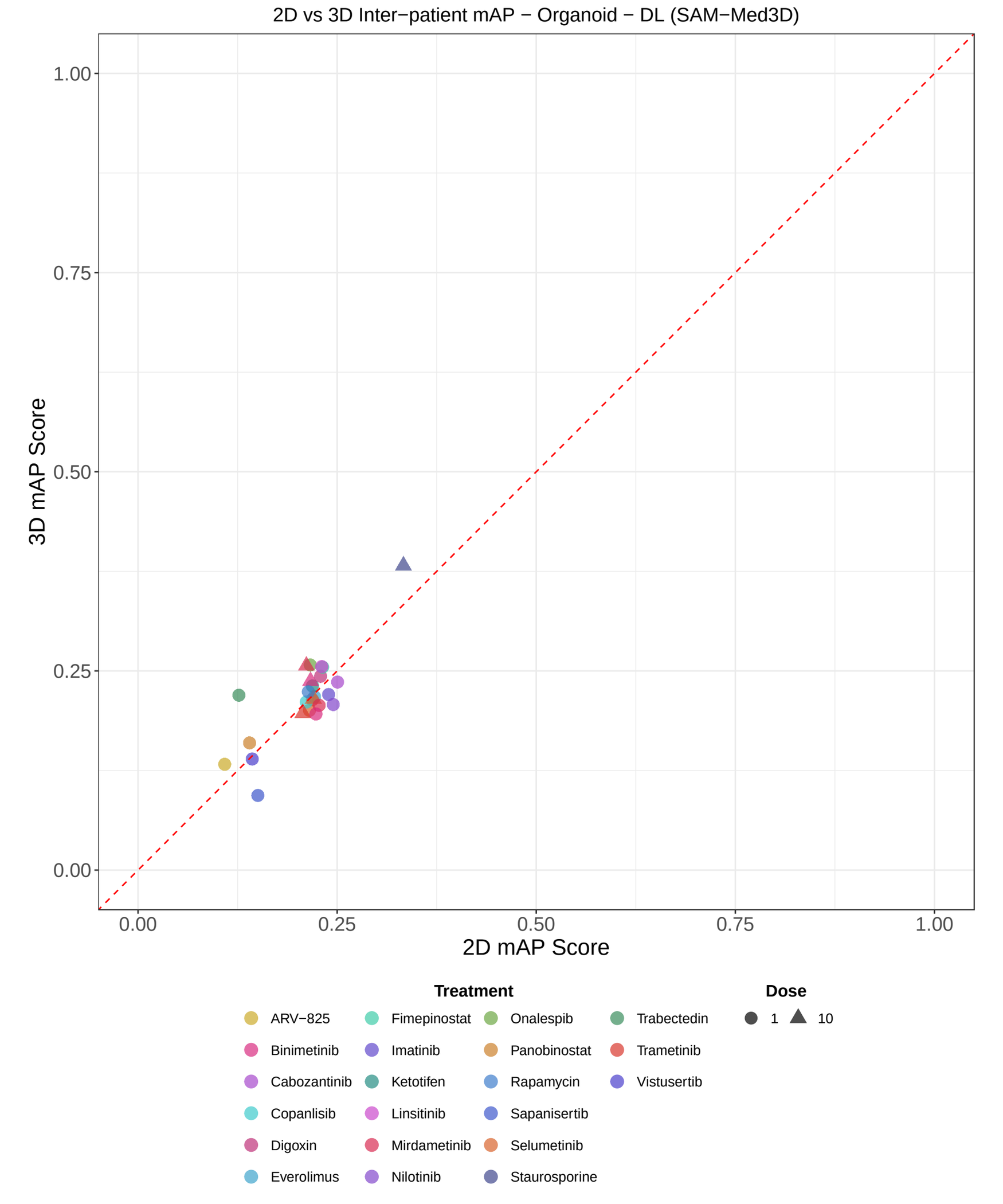

Inter−patient mAP − 3D Single−cell − Nucleocentric DL (SAM−Med3D)
−log10(p−value)
6
4
2
0
0.00
0.25
0.50
0.75
1.00
Mean Average Precision
Treatment
ARV−825
Binimetinib
Cabozantinib
Copanlisib
Digoxin
Everolimus
Fimepinostat
Imatinib
Ketotifen
Linsitinib
Mirdametinib
Nilotinib
Onalespib
Panobinostat
Rapamycin
Sapanisertib
Selumetinib
Staurosporine
Trabectedin
Trametinib
Vistusertib
Dose
1
10

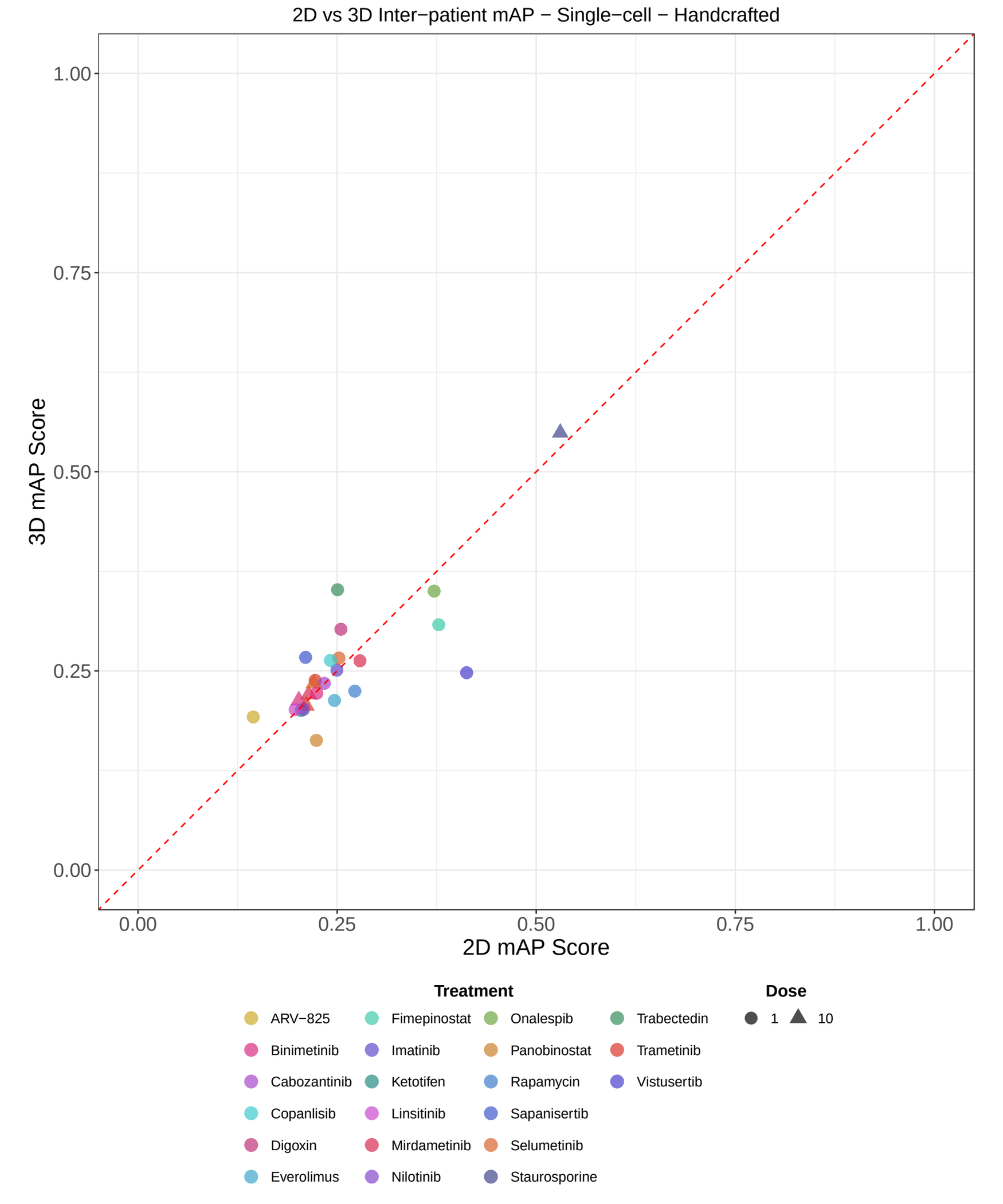

Inter−patient mAP vs Distance − 3D Single−cell − Nucleocentric DL (SAM−Med3D)
Cosine Distance
0.75
0.70
0.65
0.60
0.00
0.25
0.50
0.75
1.00
Mean Average Precision
Treatment
ARV−825
Binimetinib
Cabozantinib
Copanlisib
Digoxin
Everolimus
Fimepinostat
Imatinib
Ketotifen
Linsitinib
Mirdametinib
Nilotinib
Onalespib
Panobinostat
Rapamycin
Sapanisertib
Selumetinib
Staurosporine
Trabectedin
Trametinib
Vistusertib
Dose
1
10

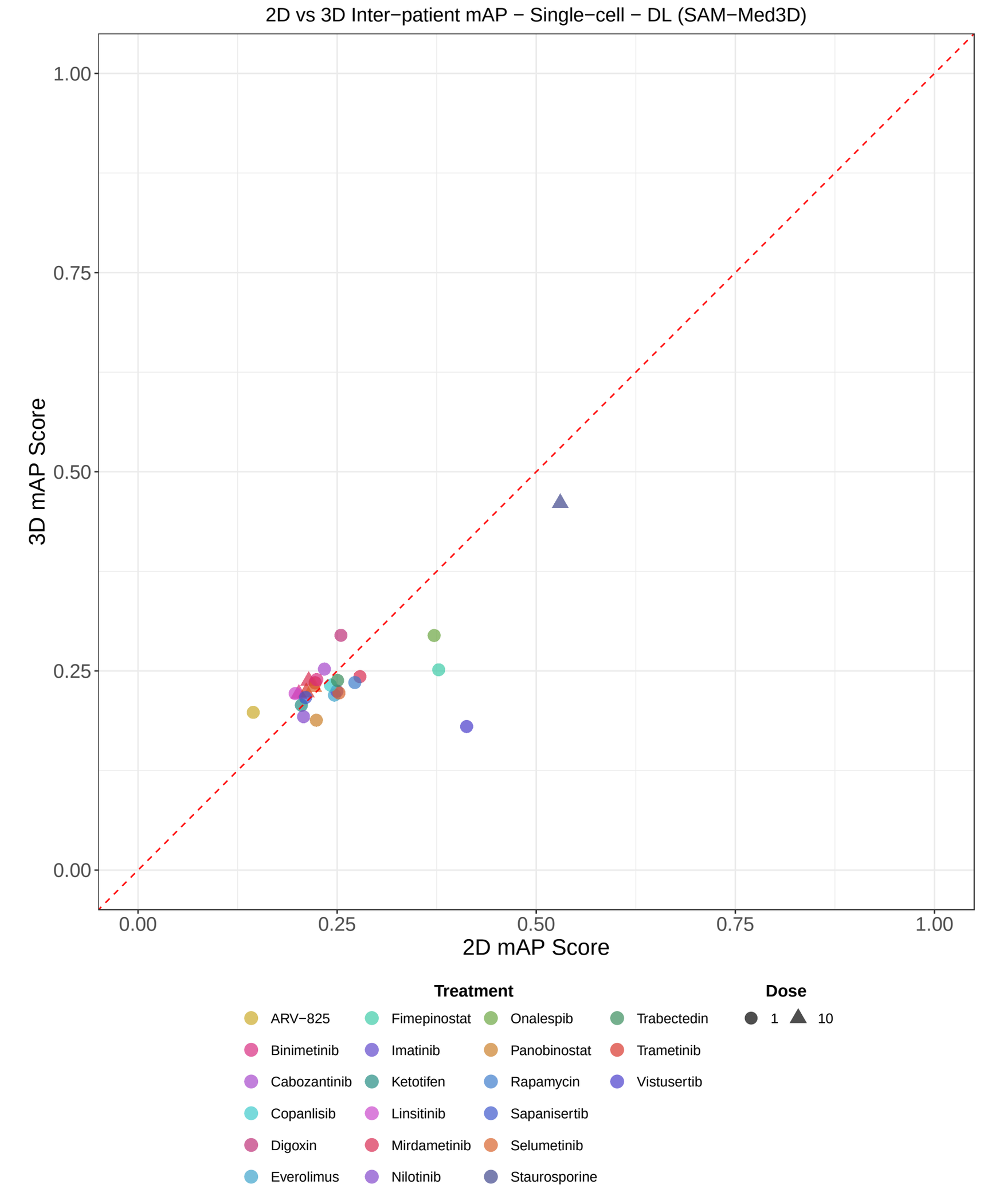

Inter−patient mAP − 3D Single−cell − Nucleocentric DL (MorphEM)
−log10(p−value)
6
4
2
0.00
0.25
0.50
0.75
1.00
Mean Average Precision
Treatment
ARV−825
Binimetinib
Cabozantinib
Copanlisib
Digoxin
Everolimus
Fimepinostat
Imatinib
Ketotifen
Linsitinib
Mirdametinib
Nilotinib
Onalespib
Panobinostat
Rapamycin
Sapanisertib
Selumetinib
Staurosporine
Trabectedin
Trametinib
Vistusertib
Dose
1
10

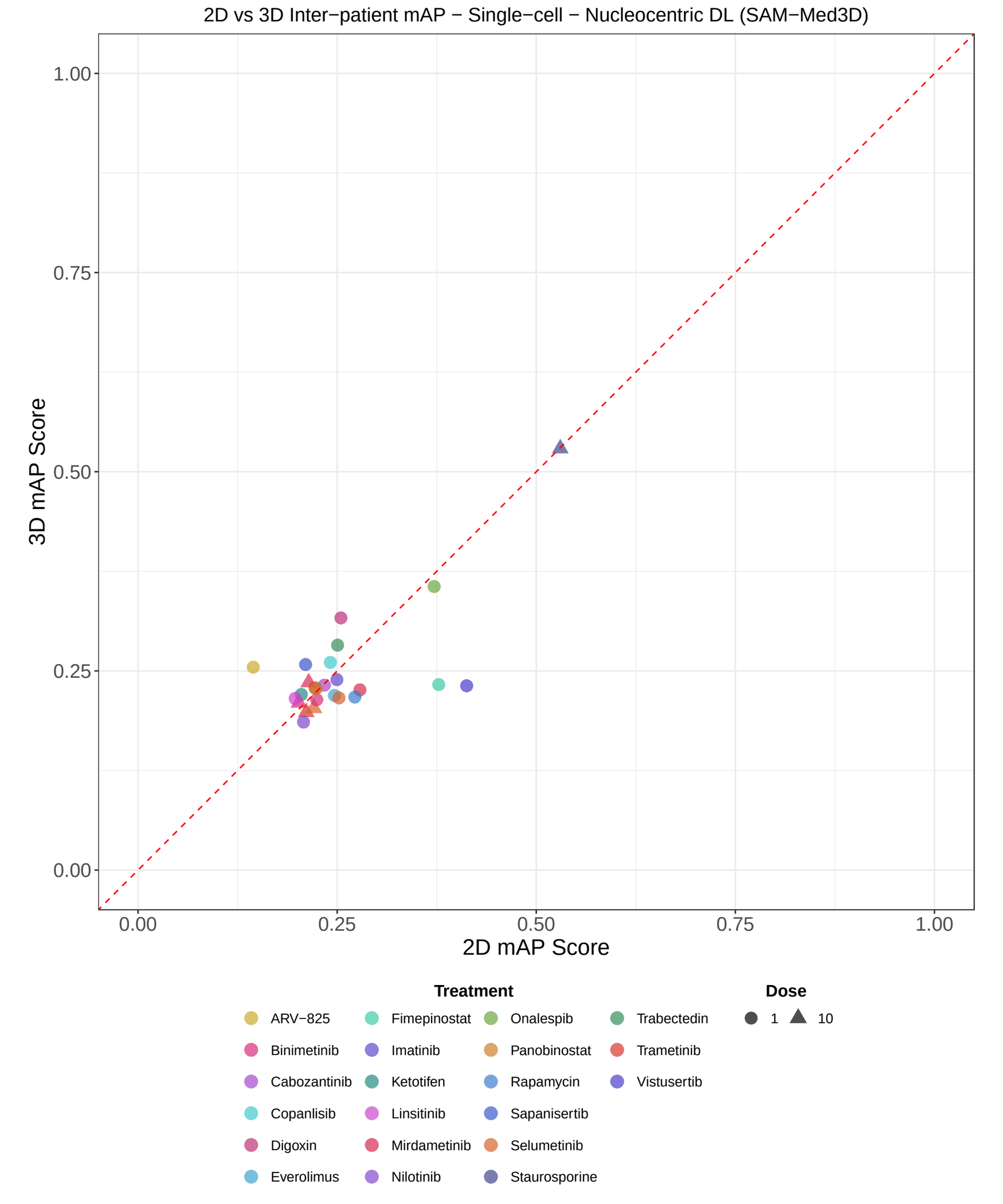

Inter−patient mAP vs Distance − 3D Single−cell − Nucleocentric DL (MorphEM)
Cosine Distance
1.00
0.99
0.98
0.97
0.00
0.25
0.50
0.75
1.00
Mean Average Precision
Treatment
ARV−825
Binimetinib
Cabozantinib
Copanlisib
Digoxin
Everolimus
Fimepinostat
Imatinib
Ketotifen
Linsitinib
Mirdametinib
Nilotinib
Onalespib
Panobinostat
Rapamycin
Sapanisertib
Selumetinib
Staurosporine
Trabectedin
Trametinib
Vistusertib
Dose
1
10

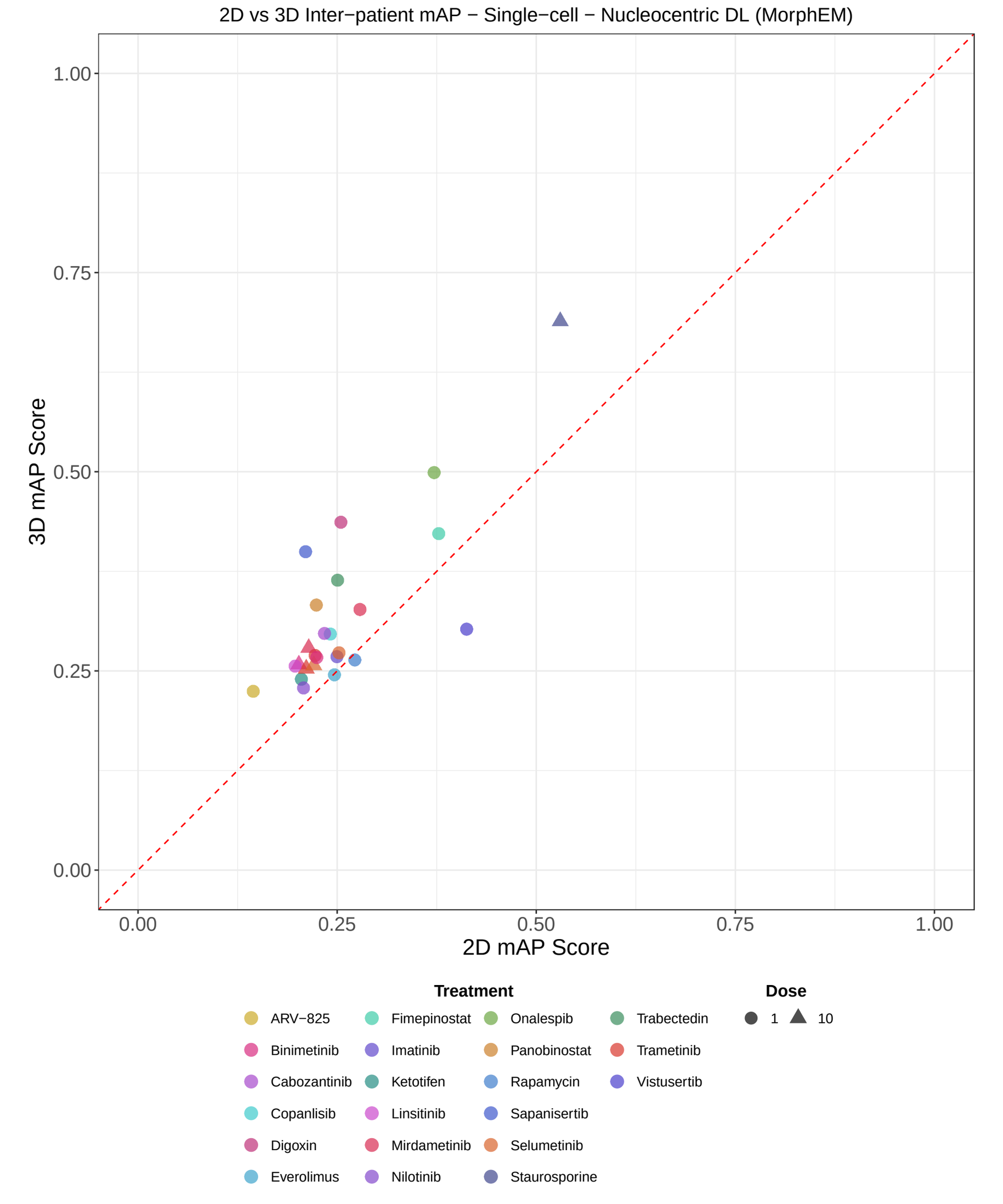

2D vs 3D Inter−patient mAP − Organoid − Handcrafted
3D mAP Score
2D mAP Score
1.00
0.75
0.50
0.25
0.00
0.00
0.25
0.50
0.75
1.00
Treatment
ARV−825
Binimetinib
Cabozantinib
Copanlisib
Digoxin
Everolimus
Fimepinostat
Imatinib
Ketotifen
Linsitinib
Mirdametinib
Nilotinib
Onalespib
Panobinostat
Rapamycin
Sapanisertib
Selumetinib
Staurosporine
Trabectedin
Trametinib
Vistusertib
Dose
1
10

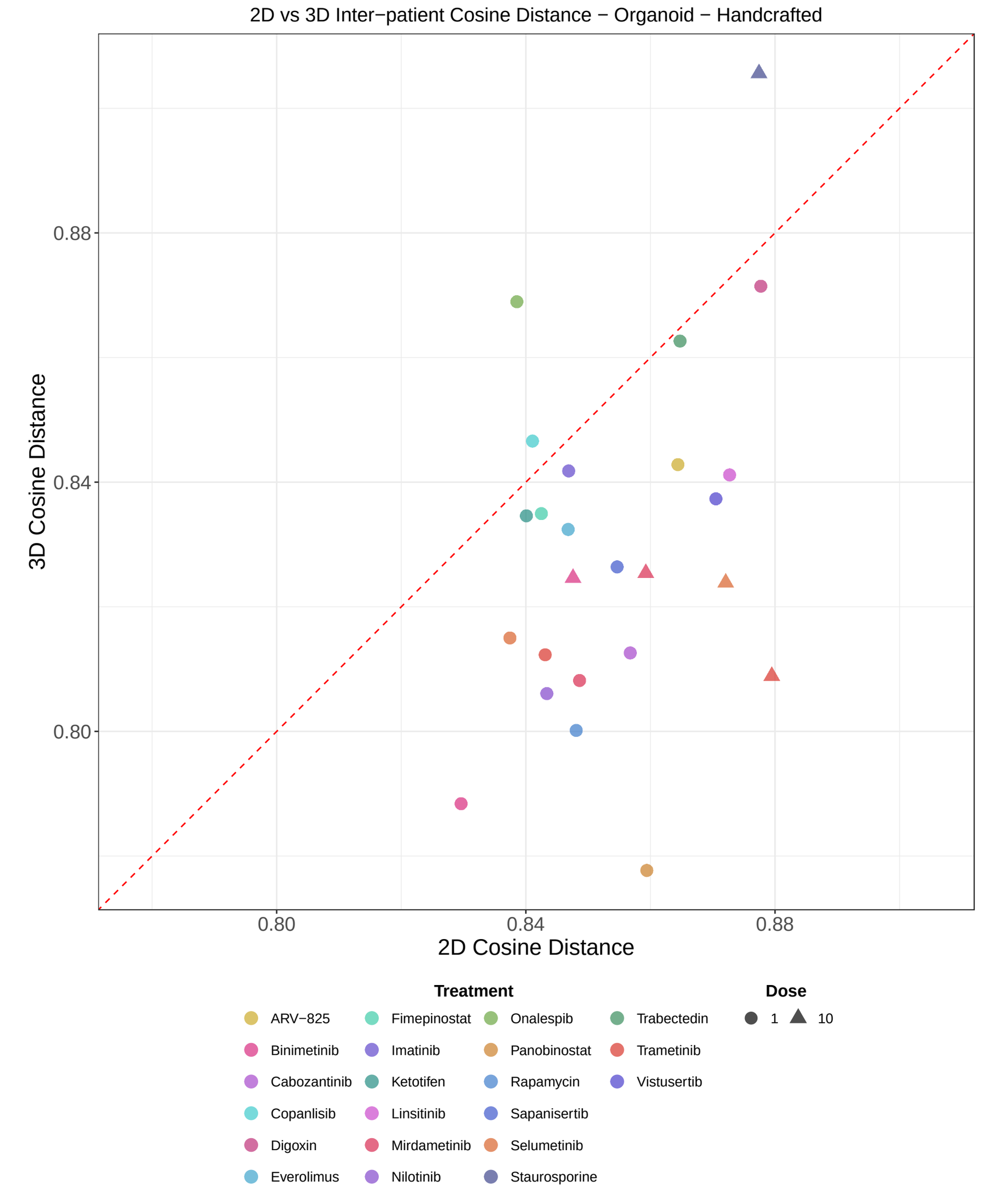

2D vs 3D Inter−patient mAP − Organoid − DL (SAM−Med3D)
3D mAP Score
2D mAP Score
0.00
0.25
0.50
0.75
1.00
Treatment
ARV−825
Binimetinib
Cabozantinib
Copanlisib
Digoxin
Everolimus
Fimepinostat
Imatinib
Ketotifen
Linsitinib
Mirdametinib
Nilotinib
Onalespib
Panobinostat
Rapamycin
Sapanisertib
Selumetinib
Staurosporine
Trabectedin
Trametinib
Vistusertib
Dose
1
10

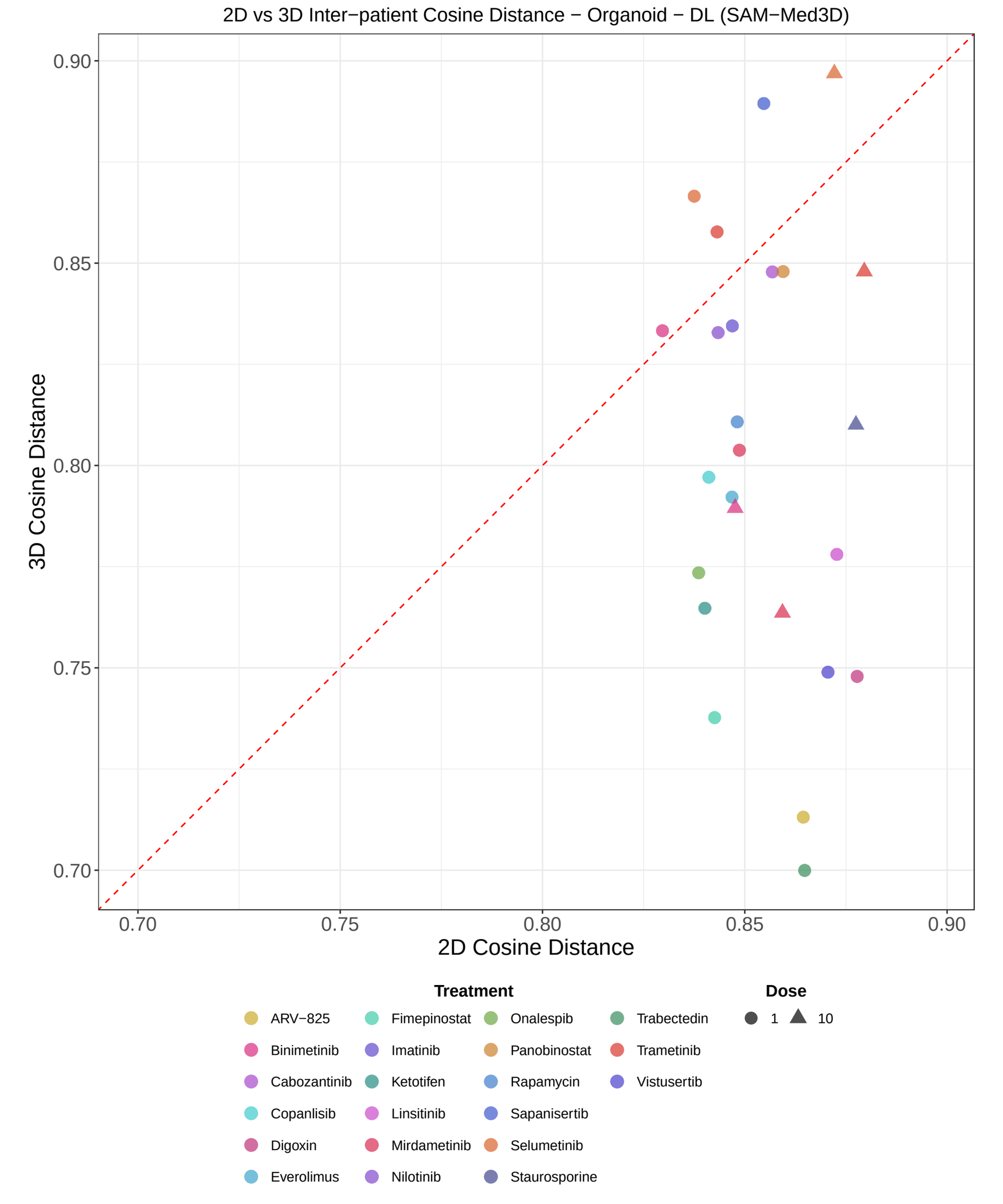

2D vs 3D Inter−patient mAP − Single−cell − Handcrafted
3D mAP Score
2D mAP Score
0.00
0.25
0.50
0.75
1.00
Treatment
ARV−825
Binimetinib
Cabozantinib
Copanlisib
Digoxin
Everolimus
Fimepinostat
Imatinib
Ketotifen
Linsitinib
Mirdametinib
Nilotinib
Onalespib
Panobinostat
Rapamycin
Sapanisertib
Selumetinib
Staurosporine
Trabectedin
Trametinib
Vistusertib
Dose
1
10

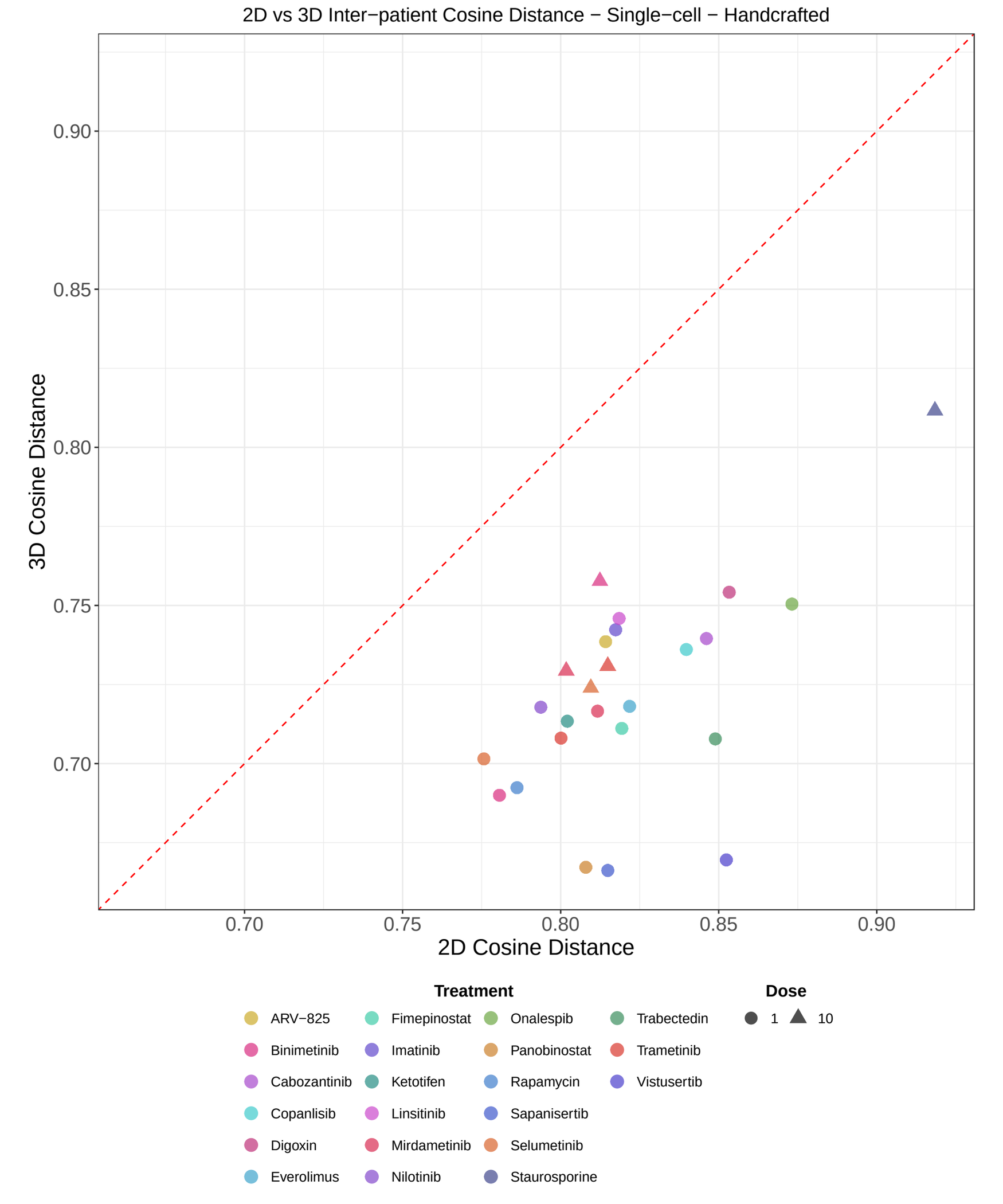

2D vs 3D Inter−patient mAP − Single−cell − DL (SAM−Med3D)
3D mAP Score
2D mAP Score
0.00
0.25
0.50
0.75
1.00
Treatment
ARV−825
Binimetinib
Cabozantinib
Copanlisib
Digoxin
Everolimus
Fimepinostat
Imatinib
Ketotifen
Linsitinib
Mirdametinib
Nilotinib
Onalespib
Panobinostat
Rapamycin
Sapanisertib
Selumetinib
Staurosporine
Trabectedin
Trametinib
Vistusertib
Dose
1
10

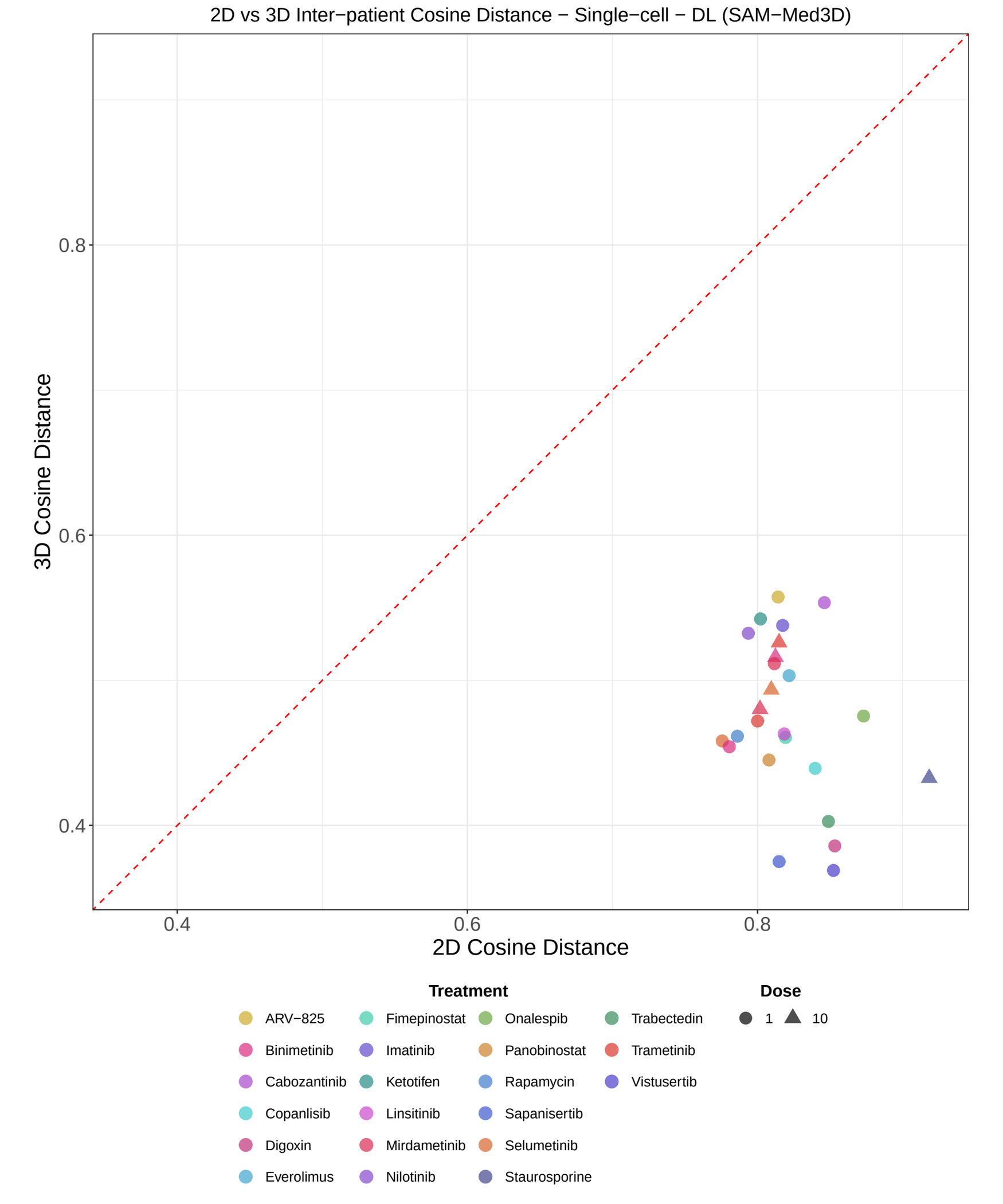

2D vs 3D Inter−patient mAP − Single−cell − Nucleocentric DL (SAM−Med3D)
3D mAP Score
2D mAP Score
0.00
0.25
0.50
0.75
1.00
Treatment
ARV−825
Binimetinib
Cabozantinib
Copanlisib
Digoxin
Everolimus
Fimepinostat
Imatinib
Ketotifen
Linsitinib
Mirdametinib
Nilotinib
Onalespib
Panobinostat
Rapamycin
Sapanisertib
Selumetinib
Staurosporine
Trabectedin
Trametinib
Vistusertib
Dose
1
10

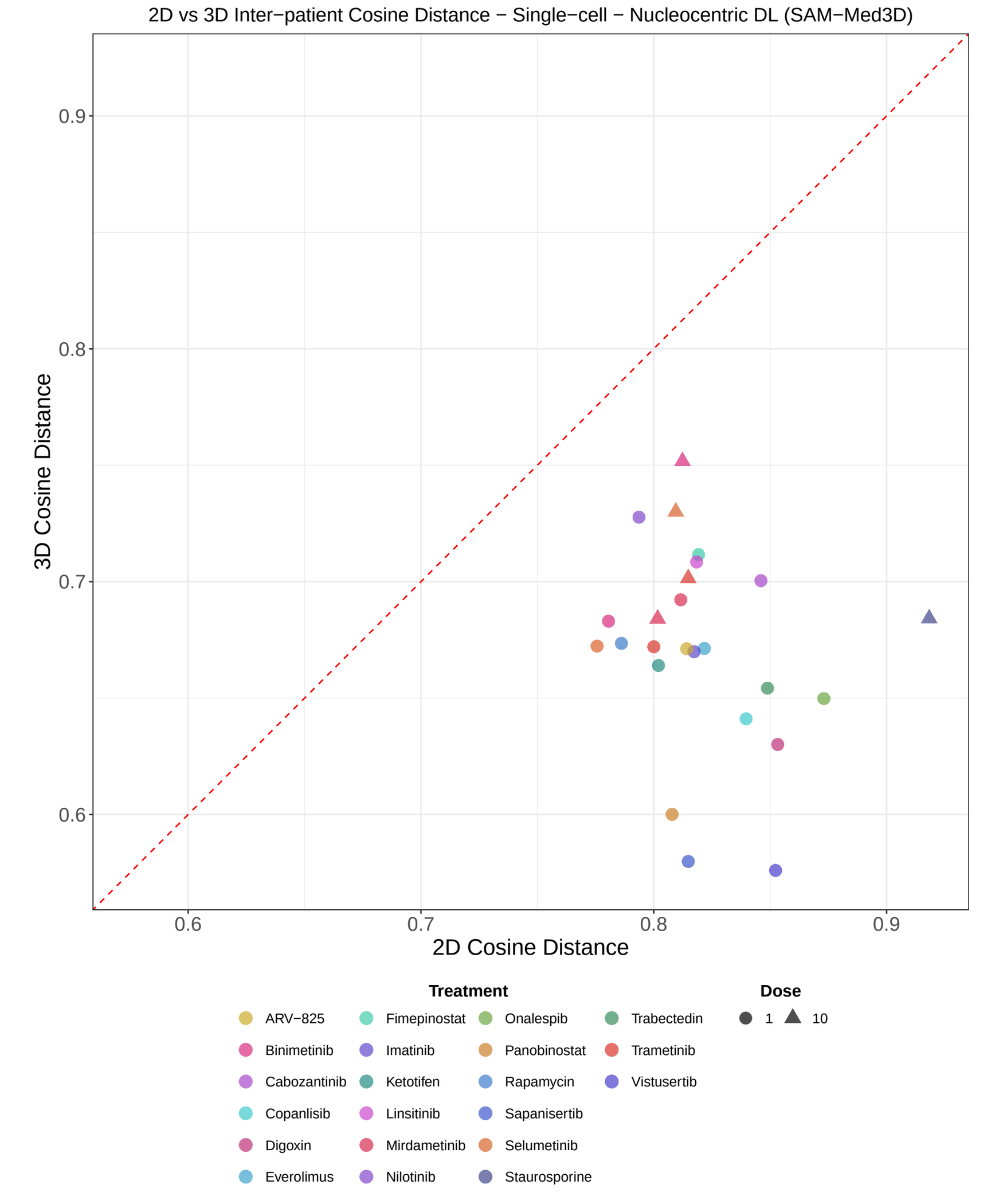

2D vs 3D Inter−patient mAP − Single−cell − Nucleocentric DL (MorphEM)
3D mAP Score
2D mAP Score
1.00
0.75
0.50
0.25
0.00
0.00
0.25
0.50
0.75
1.00
Treatment
ARV−825
Binimetinib
Cabozantinib
Copanlisib
Digoxin
Everolimus
Fimepinostat
Imatinib
Ketotifen
Linsitinib
Mirdametinib
Nilotinib
Onalespib
Panobinostat
Rapamycin
Sapanisertib
Selumetinib
Staurosporine
Trabectedin
Trametinib
Vistusertib
Dose
1
10

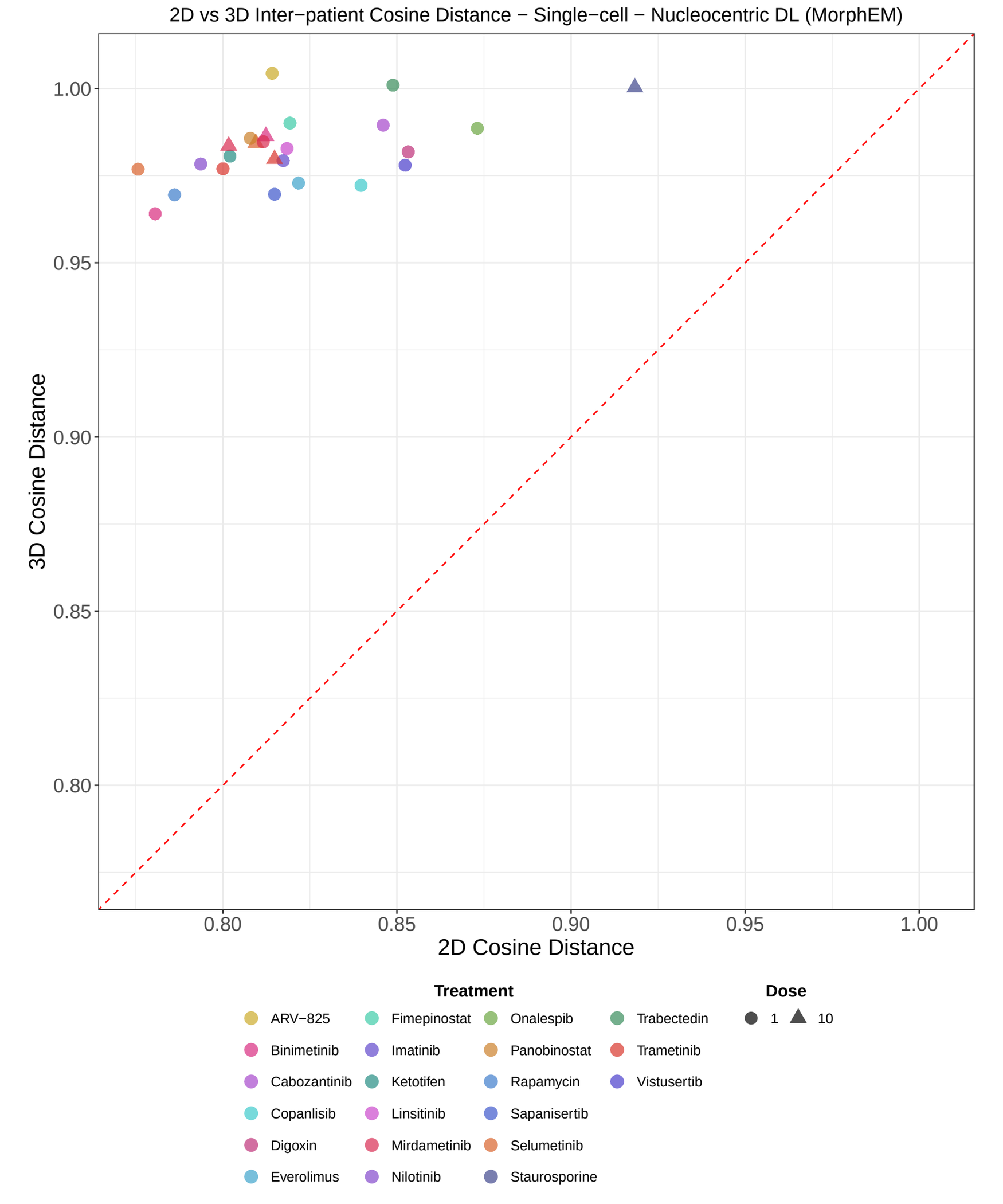

2D vs 3D Inter−patient Cosine Distance − Organoid − Handcrafted
3D Cosine Distance
0.88
0.84
0.80
0.80
0.84
0.88
2D Cosine Distance
Treatment
ARV−825
Binimetinib
Cabozantinib
Copanlisib
Digoxin
Everolimus
Fimepinostat
Imatinib
Ketotifen
Linsitinib
Mirdametinib
Nilotinib
Onalespib
Panobinostat
Rapamycin
Sapanisertib
Selumetinib
Staurosporine
Trabectedin
Trametinib
Vistusertib
Dose
1
10


2D vs 3D Inter−patient Cosine Distance − Organoid − DL (SAM−Med3D)
3D Cosine Distance
0.90
0.85
0.80
0.75
0.70
0.70
0.75
0.80
0.85
0.90
2D Cosine Distance
Treatment
ARV−825
Binimetinib
Cabozantinib
Copanlisib
Digoxin
Everolimus
Fimepinostat
Imatinib
Ketotifen
Linsitinib
Mirdametinib
Nilotinib
Onalespib
Panobinostat
Rapamycin
Sapanisertib
Selumetinib
Staurosporine
Trabectedin
Trametinib
Vistusertib
Dose
1
10


2D vs 3D Inter−patient Cosine Distance − Single−cell − Handcrafted
3D Cosine Distance
0.90
0.85
0.80
0.75
0.70
0.70
0.75
0.80
0.85
0.90
2D Cosine Distance
Treatment
ARV−825
Binimetinib
Cabozantinib
Copanlisib
Digoxin
Everolimus
Fimepinostat
Imatinib
Ketotifen
Linsitinib
Mirdametinib
Nilotinib
Onalespib
Panobinostat
Rapamycin
Sapanisertib
Selumetinib
Staurosporine
Trabectedin
Trametinib
Vistusertib
Dose
1
10


2D vs 3D Inter−patient Cosine Distance − Single−cell − DL (SAM−Med3D)
3D Cosine Distance
0.8
0.6
0.4
0.4
0.6
0.8
2D Cosine Distance
Treatment
ARV−825
Binimetinib
Cabozantinib
Copanlisib
Digoxin
Everolimus
Fimepinostat
Imatinib
Ketotifen
Linsitinib
Mirdametinib
Nilotinib
Onalespib
Panobinostat
Rapamycin
Sapanisertib
Selumetinib
Staurosporine
Trabectedin
Trametinib
Vistusertib
Dose
1
10


2D vs 3D Inter−patient Cosine Distance − Single−cell − Nucleocentric DL (SAM−Med3D)
3D Cosine Distance
2D Cosine Distance
0.6
0.7
0.8
0.9
Treatment
ARV−825
Binimetinib
Cabozantinib
Copanlisib
Digoxin
Everolimus
Fimepinostat
Imatinib
Ketotifen
Linsitinib
Mirdametinib
Nilotinib
Onalespib
Panobinostat
Rapamycin
Sapanisertib
Selumetinib
Staurosporine
Trabectedin
Trametinib
Vistusertib
Dose
1
10


2D vs 3D Inter−patient Cosine Distance − Single−cell − Nucleocentric DL (MorphEM)
3D Cosine Distance
1.00
0.95
0.90
0.85
0.80
0.80
0.85
0.90
0.95
1.00
2D Cosine Distance
Treatment
ARV−825
Binimetinib
Cabozantinib
Copanlisib
Digoxin
Everolimus
Fimepinostat
Imatinib
Ketotifen
Linsitinib
Mirdametinib
Nilotinib
Onalespib
Panobinostat
Rapamycin
Sapanisertib
Selumetinib
Staurosporine
Trabectedin
Trametinib
Vistusertib
Dose
1
10